Lean Years for Productivity but Seeds of Innovation: Effects of Pesticide Regulation in French Agriculture*

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Abstract

Economists have long debated the impact of regulation on productivity. Identifying this impact is challenging because many confounding factors typically affect a firm's production function. This paper examines it in a setting with a simple production function: French crop farming. Specifically, I study how increasingly restrictive pesticide bans have affected farm productivity and agrochemical innovation. To this end, I combine French farm-level balance sheets and crop protection records with the registry of pesticide authorizations. I find that bans on inputs without substitutes are the most costly for farms, but also induce the most innovation, including entirely new active molecules. Regulation therefore involves a trade-off between short-term productivity losses and medium-term technological progress.

JEL Classifications: L51, O31, Q12, Q58.

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1 Introduction

The relationship between regulation and productivity is at the heart of a long-standing debate in economics. Regulation is often justified as a means of correcting negative externalities, such as environmental damage, but it also imposes constraints on firms and producers, which may hinder productivity growth. Regulation can, however, also create new market opportunities, either by banning existing products or by precisely defining new markets. By raising the expected returns to innovation, a central mechanism in endogenous growth models (Romer, 1990; Aghion and Howitt, 1992), it can then act as a catalyst for technological progress.

This debate is central to current European policy discussions. In 2023, the European Union adopted a regulation aimed at ending the sale of new cars with internal combustion engines from 2035.[1] Pesticide regulation, the focus of this paper, is equally contested, as illustrated by the debates over the renewal of the approval of glyphosate in 2023 and over the ban on neonicotinoid insecticides.[2] In the short run, such bans deprive farmers of key crop protection products and force them to rely on costlier or less effective alternatives, with a risk of productivity losses. In the medium run, they may strengthen the incentives of agrochemical companies to develop new solutions, with potential gains in productivity. The current stance of the European Commission is that pesticides should be banned only when alternatives exist. As the European Commissioner for Agriculture, Christophe Hansen, put it: "We should not ban pesticides if no alternatives are available."[3] Is this the optimal policy response? This paper shows that the answer involves a trade-off. Bans on products without substitutes are the most costly for the farms that rely on them, but they are also those that induce the most innovation.

Richness of French agricultural data. I study this question in a setting with a simple production function: French crop farming. In this setting, crops are distinct product lines, and regulation targets specific inputs for specific products. Recently released French agricultural data offer an unprecedented level of detail for this purpose. Unlike standard firm-level datasets, which provide limited information on inputs, they describe both the inputs that farms use and the outputs they produce. This richness stems from exhaustive administrative registers and detailed surveys, which make agricultural data an invaluable resource for studying economic mechanisms that are hard to identify in standard firm-level data. I build a novel dataset that combines balance-sheet data on the universe of French farms, the Graphical Land Parcel Register, which records the crop grown on every parcel each year, and the Field Crop Cultural Practices Survey, which records the crop protection products applied on the surveyed parcels. I complement these data with the Farm Accountancy Data Network, which sep-

[1] Regulation (EU) 2023/851 sets a 100% reduction target for the CO_2 emissions of new cars and vans from 2035. On December 16, 2025, the European Commission proposed to soften this target by lowering it to 90%.

[2] Commission Implementing Regulation (EU) 2023/2660 renewed the approval of glyphosate until December 15, 2033.

[3] Interview in *Le Monde*, February 19, 2025.

arates yields from prices for each crop. Finally, I use the E-Phy registry of crop protection products, which records the authorization status and the authorized uses of every product, to identify regulatory events and to measure innovation.

A ban without substitutes. I first study a typical case of a ban on an input without substitutes: the 2018 French ban on neonicotinoid insecticides. Sugar beet growers relied on neonicotinoid seed coatings to protect their crops against the aphids that transmit beet yellows virus: 96% of sugar beet farms used them in 2017, and no effective alternative was available at the time of the ban. The European Union is the world's leading producer of beet sugar, with around half of world production,[4] and in the counties of northern France where it is concentrated, sugar beet accounts for up to 12% of agricultural land. A key challenge is that farms may have specialized in sugar beet for reasons that also affect their productivity. I address it by exploiting the European sugar quota system, which, for five decades, froze the allocation of sugar beet cultivation across farms, so that within a given locality, exposure to the ban was inherited rather than chosen. Comparing farms with different pre-ban reliance on sugar beet within the same city, I find that the more exposed farms were to the ban, the more their land productivity declined. Between 2018 and 2022, each additional percentage point of pre-ban sugar beet land share reduced revenue land productivity by about 0.3%, which corresponds to a decline of about 5% for highly exposed farms relative to less exposed farms. Value-added land productivity declined about three times as much, since spending on intermediate inputs did not decline, and spending on crop protection products even increased in the first full season under the ban. To understand this decline, I decompose it into three channels. Lower sugar beet yields account for about a third of the decline, and the persistent reorganization of the crop mix induced by the ban for about a fifth. The remainder reflects relative price movements on world commodity markets. The lower activity of exposed farms also reduced their labor demand: their wage bill declined. Finally, beyond its costs for the farms it targeted, the ban generated positive environmental spillovers. By protecting pollinators, it raised the land productivity of neighboring orchards, whose yields depend on pollination, although its effect on neighboring farms as a whole is a precisely estimated zero.

Availability of substitutes and cost of bans. The neonicotinoid ban is an illustrative but specific case: it concerned a single crop, and no substitute was available. I then extend the analysis to all bans on crop protection products that affected French field crops in 2017 and 2018. A farm is exposed if, in 2017, it used on an economically relevant crop a product whose use on this crop was banned in 2017 or 2018. On average, bans reduce the land productivity of exposed farms by about 3% relative to non-exposed farms of the same county, about five years after the ban. This average masks large differences, which depend on the availability of substitutes for the banned products. When no substitute is available, land productivity declines by about 10%, a pattern close to that of the neonicotinoid ban.

[4] European Commission, Sugar.

When substitutes exist but are scarce, the decline is halved, and when they are abundant, bans have no effect. Farm practices also matter. When substitutes are scarce, exposed farms that had adopted practices to reduce their dependence on pesticides before the bans experienced no loss. When no substitute is available, by contrast, these practices do not soften the effect of the bans. From a static perspective, these results support the principle of not banning a product without substitutes.

How bans induce innovation. This static view, however, takes the set of available technologies as given. In endogenous growth models, innovation is driven by the prospect of monopoly rents. A ban that leaves a use without any authorized product creates a void that a new product can fill with little competition, whereas a ban on a product with many close substitutes leaves the market competitive and offers weaker incentives to innovate (Aghion et al., 2005). I test this prediction at the level of uses, defined as crop–target pairs (e.g., aphid control in sugar beet), which correspond to the markets in which crop protection products compete. I compare uses that lose all their authorized products in the largest ban of their history with uses that keep at least one alternative authorized by the regulator. Bans that leave a use without substitutes increase the probability of innovation for this use by about 5 percentage points six to eight years later. This effect represents about 40% of the average annual probability of innovation. Agrochemical firms respond both by adapting substances already authorized for other uses and by introducing entirely new molecules. This innovation, however, takes time: eight years after the ban, most uses left without substitutes still have no authorized active substance, and farmers face constrained pest control options in the meantime.

Implications for regulation. The effect of bans on innovation is thus the mirror image of their effect on productivity: bans on products without substitutes are the most costly for farms, but also those that induce the most innovation. Regulation therefore involves a trade-off between short-term productivity losses and medium-term technological progress. Bans on products without substitutes can be justified not only by the negative environmental externalities of these products, but also as a way to induce the development of new pest control solutions. They should then be accompanied by support targeted at the losses caused by the regulation, and by the promotion of practices that reduce the dependence of farms on chemical inputs. Regulation should also be predictable and credible, both because farms reorganize their production in response to bans and because agrochemical firms invest more in research when they expect the demand for substitutes to last.

Related literature. This paper contributes to four strands of literature.

First, it contributes to the literature on the effects of regulation on firm performance. A large body of work studies the effects of environmental regulation on industrial activity, productivity, and workers (Berman and Bui, 2001; Greenstone, 2002; Greenstone et al., 2012; Walker, 2013), and other studies document the effects of size-dependent labor regulations (Garicano et al., 2016) and mandatory profit-sharing on French firms (Nimier-David et al., 2023). Identifying these effects is difficult, be-

cause regulation rarely targets an input that is easily measurable in firm-level data, and because many factors shape a firm's production function at the same time. I study regulations that remove a specific input for a specific product line, in a setting where inputs, outputs, and technologies are observed in detail. This makes it possible to quantify firm-level mechanisms with unusual precision, and in particular to separate the direct effect of a regulation on the product line it targets from the induced reorganization of production across product lines.

Second, it contributes to the literature on the effects of regulation on innovation. Porter and van der Linde (1995) argue that well-designed environmental regulation can trigger innovations that partly or fully offset the costs of complying with it. The literature on directed technical change formalizes how policy can redirect innovation toward clean technologies (Acemoglu et al., 2012), and empirical work shows that energy prices, fuel prices, and carbon markets redirect innovation accordingly (Popp, 2002; Aghion et al., 2016; Calel and Dechezleprêtre, 2016). More generally, innovation responds to the expected size of the market (Acemoglu and Linn, 2004; Finkelstein, 2004). Bans can induce innovation as well: Dugoua (2025) shows that the phase-out of chlorofluorocarbons under the Montreal Protocol induced innovation in their substitutes. Regulation can, however, also discourage innovation when it raises the cost of firm growth (Aghion et al., 2023). I show that bans induce innovation, including the introduction of entirely new molecules, and that the strength of this response depends on the availability of substitutes, which also determines the cost of bans for the firms that rely on the banned inputs. This paper thus links the static costs of regulation and its dynamic benefits within a single setting.

Third, it contributes to the literature on the drivers of agricultural productivity. A large literature documents cross-country differences in agricultural productivity and their determinants (Gollin et al., 2014a,b; Chen, 2017; Adamopoulos and Restuccia, 2022; Moscona and Sastry, 2022; Boppart et al., 2023). Another strand studies the diffusion of new agricultural technologies, such as hybrid crops (Griliches, 1957) and tractors (Manuelli and Seshadri, 2014; Gross, 2018), the effect of land consolidation on farm productivity (Chabé-Ferret and Enrich, 2021), and the structural transformation that new agricultural technologies can trigger (Bustos et al., 2016). I document, at the micro level and in a developed economy, the converse effect: how the withdrawal of an existing technology affects farm productivity, and through which channels.

Fourth, it contributes to the literature on agriculture and the environment. Recent work studies how agricultural innovation responds to climate change (Moscona and Sastry, 2023) and the costs of adapting to it (Du Puy and Shrader, 2024). Regarding pesticides, Dias et al. (2023) and Frank (2024) document their costs for human health, Gaigné et al. (2025) show that the pesticide content of European consumption is more intensive than that of European production, which raises concerns about the role of trade in undermining pesticide-reduction targets, and Frank et al. (2026) study the effects of the

French neonicotinoid ban on biodiversity. I provide evidence on the productivity effects of a large-scale environmental regulation, both on the farms it targets and, through its environmental spillovers, on neighboring farms.

Outline. The remainder of the paper is organized as follows. Section 2 describes the data. Section 3 examines a case of regulation without alternatives: the French neonicotinoid ban. Section 4 extends the analysis to all pesticide bans affecting French field crops and examines the role of substitutes. Section 5 studies the effect of bans on innovation. Section 6 concludes.

2 Data

To analyze the effects of pesticide regulation on farm productivity and innovation, I assemble a unique and novel combination of datasets on French agriculture, several of which have only recently been made available to researchers. This combination makes it possible to study with precision the link between regulation and productivity. Information on farms' economic performance is obtained from the Agricultural Benefits Database, which contains exhaustive balance-sheet data. The Graphical Land Parcel Register locates each parcel geographically every year and records the crop grown on it, thereby describing the structure of production of the universe of farms. These exhaustive data on outputs and inputs are usefully complemented by thematic surveys: the Field Crop Cultural Practices Survey provides detailed information on the technologies used by farms, including the plant protection products applied to each parcel, while the Farm Accountancy Data Network records output by crop, which makes it possible to separate prices from yields. The Agricultural Census records farm land use as far back as 1988, decades before the regulatory changes studied. Finally, the E-Phy database, the registry of plant protection products, records their authorization status by year, which I use to define regulatory events and to measure innovation. In this section, I describe each data source and how it is used; the samples and descriptive statistics are presented with each empirical design.

Agricultural Benefits Database ("BA"). This database covers the entire population of farms in France's agricultural sector from 1995 to 2022, representing approximately 220,000 farms per year. While the Ficus-Fare database, which details balance-sheet records for non-agricultural business firms, serves as the benchmark for firm-level studies in France, its counterpart in the agricultural sector, the Agricultural Benefits Database ("BA"), provides detailed information for each farm. The recent availability of this database offers new perspectives for research on these issues, as previous data sources were limited to thematic surveys, and the relatively small number of farms surveyed made it difficult, if not impossible, to match data across multiple sources. I use these balance-sheet

data to construct farm-level measures of production (total production, crop production, and value added), land productivity, tangible assets (stock of machines), and intermediate consumption by type.

Graphical Land Parcel Register. This database covers all French land parcels receiving European subsidies from the Common Agricultural Policy (CAP) between 2015 and 2022, providing highly detailed annual information on parcel size, the crop grown, the identifier of the farm cultivating the parcel, and the geographic boundaries of the parcel. I use this information to construct precise measures of farm land use allocation across different crop types, allowing me to measure each farm's pre-ban exposure to sugar beet cultivation and to analyze potential changes in the production function following shocks such as pesticide bans. The geographic boundaries of parcels also allow me to compute distances between parcels and, for each farm, the area devoted to each crop in its neighborhood. I use these measures to estimate the spillover effects of pesticides ban on neighboring farms.

E-Phy database. The E-Phy database provides access to detailed information on plant protection products covered by a marketing authorization by the *French Agency for Food, Environmental and Occupational Health & Safety (ANSES)*. The dataset files cover approximately 15,000 authorized and withdrawn products since the 1960s. They include the crop protection product identifier, the date of marketing authorization, the current authorization status (approved or banned), the potential ban date, the active substance content, as well as product uses. Product uses cover two dimensions: the type of crop on which the product can be applied, and the target of the product (e.g., insects, fungi). I use this information to define pesticide ban events. Aggregating products by use, I also build annual panels of uses (e.g., aphid control in sugar beet) recording the products and active substances authorized each year. These panels allow me to count the substitutes available for each banned product and to measure innovation at the crop–target level.

Field Crop Cultural Practices Survey. The Field Crop Cultural Practices Survey is conducted at the land parcel level for the years 2001, 2006, 2011, 2017, and 2021, covering approximately 20,000 land parcels each year. For each land parcel, the survey provides detailed information on the crop cultivated, the plant protection products applied during cultivation, and the agricultural practices employed (e.g., tillage). This information, in particular, allows for the identification of farms that would be affected by a ban on a specific plant protection product or active substance.

Farm Accountancy Data Network (FADN). The Farm Accountancy Data Network (FADN) is a survey conducted annually in the member states of the European Union. In France, data are available from 1968 to 2023, covering approximately 7,000 farms each year since 1988. Unlike the BA, which

only reports aggregate nominal revenues, it records for each crop the area cultivated, the quantity produced, and the value of production, making it possible to compute crop-specific yields, prices, and revenues per hectare at the farm level. However, it only covers around 2% to 3% of all farms each year. Moreover, matching this subsample with the *Field Crop Cultural Practices Survey* results in an even more limited overlap, making it difficult to study the effects of regulatory shocks or policy changes at the farm level. I therefore use this dataset as a secondary data source, to decompose the effect of the neonicotinoid ban on land productivity into its underlying mechanisms. To do so, I rely on the subsample of farms in the main sample that are also surveyed in the FADN.

Agricultural Census. The Agricultural Census is an exhaustive survey of all French farms, conducted approximately every ten years (most recently in 1988, 2000, 2010, and 2020). It records, for each farm, the area devoted to each crop, as well as information on livestock, labor, and the farm manager. I use the 1988 and 2000 censuses to measure each farm's historical exposure to sugar beet cultivation, up to three decades before the neonicotinoid ban.

3 Regulating without Alternatives: The French Neonicotinoid Ban

In this section, I study a regulatory ban that removed a key input for which no substitute was available: the 2018 ban on neonicotinoid insecticides, on which virtually all French sugar beet growers relied. I show that the revenue land productivity of farms highly exposed to the ban declines by about 5% relative to less exposed farms on average over 2018–2022, which corresponds to a decline of about 0.3% for each additional percentage point of pre-ban sugar beet land share. Value-added land productivity falls even more, as spending on intermediate inputs does not decline. Lower sugar beet yields account for about a third of the effect on land productivity and the induced reallocation of land away from sugar beet for about a fifth, the remainder reflecting relative price movements on world commodity markets. I first describe the institutional context (Section 3.1) and present a research design that exploits the European sugar quota system (Section 3.2). I then report the main results (Section 3.3), address threats to identification (Section 3.4), and decompose the effect into its underlying mechanisms (Section 3.5). Finally, Section 3.6 shows that, by protecting pollinators, the ban raised the land productivity of neighboring orchards.

3.1 Institutional Context

Neonicotinoids. Neonicotinoids are a class of neurotoxic insecticides developed by agrochemical companies in the 1980s. Their systemic action and molecular stability made them more effective than

nicotine, which they rapidly replaced during the 1990s.[5] By the mid-2010s, neonicotinoids had become the most widely used class of insecticides worldwide, accounting for more than 25% of global insecticide sales (Bass et al., 2015). A growing body of scientific evidence has, however, documented their harmful effects on pollinators and aquatic insects. In 2018, a report of the European Food Safety Authority based on a meta-analysis of 588 studies concluded that three widely used neonicotinoids pose a threat to bees (European Food Safety Authority, 2018).

Regulatory timeline. Following these findings, EU member states agreed on April 27, 2018 to ban the outdoor use of these three substances (imidacloprid, clothianidin, and thiamethoxam) from the end of 2018. France had gone further: a law adopted in August 2016 banned all neonicotinoid-based products, including seeds treated with them, from September 1, 2018. The ban was subsequently relaxed for sugar beet only. Acknowledging that "the technical alternatives currently available have proven ineffective",[6] and observing that several other member states had granted derogations to support their sugar beet growers, the French government proposed a law, adopted on December 14, 2020, that temporarily re-authorized seeds treated with two neonicotinoid active substances, imidacloprid and thiamethoxam, for sugar beet cultivation. This derogation ended in 2023, after the Court of Justice of the European Union ruled in January 2023 that member states could not grant exemptions to the EU ban on seeds treated with neonicotinoids.[7] Figure 1 summarizes this timeline.

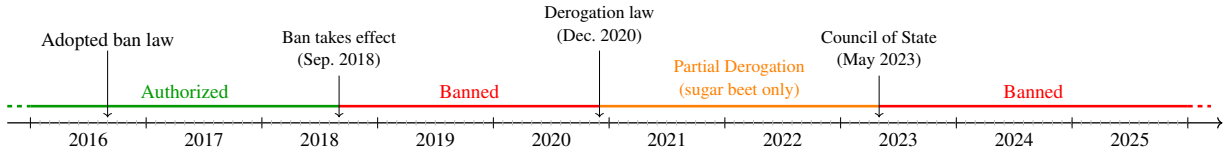


Figure 1. French Regulatory Timeline on Neonicotinoids

A critical input without substitute. Sugar beet cultivation relies heavily on neonicotinoid seed coatings, which protect the crop against aphids, the vectors of beet yellows virus. The virus causes the leaves to turn yellow, thicken, and become brittle (Appendix Figure A1a), and reduces the size of the beet (Appendix Figure A1b). In the most severe cases, yield losses can reach 50%. At the time of the ban, no effective alternative to neonicotinoids was available to control these aphids (French Agency for Food, Environmental and Occupational Health & Safety, 2018). As a result, 96% of sugar beet farms used neonicotinoids in 2017, according to the Field Crop Cultural Practices Survey.

[5] Nicotine had been used as an insecticide on fruit and horticultural crops since the 17th century. Imidacloprid, the first neonicotinoid, was discovered by Bayer in 1985 and introduced to the market in 1991.

[6] Press release of the French Ministry of Agriculture and Food, August 6, 2020.

[7] In 2025, a legislative proposal (the Duplomb Law) sought to re-authorize acetamiprid, the only neonicotinoid still approved at the EU level. The French Constitutional Council struck down this provision on environmental protection grounds.

Geography of exposure. Sugar beet is mainly cultivated in northern Europe, where the climate is most suitable. In France, cultivation is concentrated in the northern quarter of the country (Figure 2). In the counties of this region, sugar beet accounts for up to 12% of agricultural land, while it is almost absent elsewhere. The neonicotinoid ban thus primarily affected a specific region, where sugar beet represents a significant share of agricultural activity.

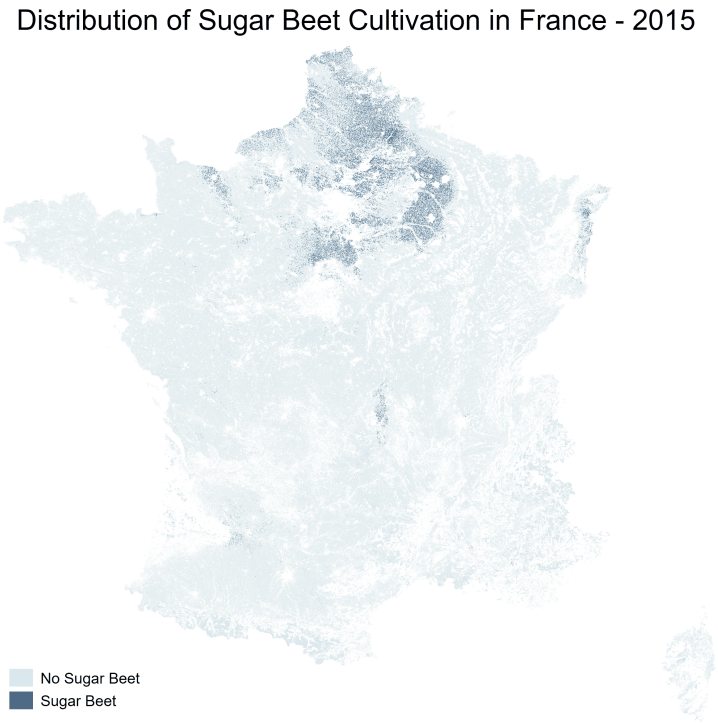


Figure 2. Sugar Beet Cultivation in 2015

Notes: This map shows the land parcels cultivated with sugar beet in 2015. Source: Graphical Land Parcel Register.

3.2 Research Design

Ideal experiment. To estimate the effect of the ban on farm productivity, I compare farms that were differentially exposed to it because of their pre-existing reliance on sugar beet. The main threat to this approach is that specialization in sugar beet may be endogenous: farms may have chosen it because of specific assets, soil quality, or expectations about future regulation, all of which could also affect their productivity. The ideal experiment would randomly assign sugar beet cultivation across farms. Such an experiment cannot be run at scale, but the European sugar quota system created a setting that approximates it.

The European sugar quota system. The common organisation of the sugar market, established in 1968, combined guaranteed minimum prices for sugar beet with production quotas.[8] Because guaranteed prices encouraged overproduction, quotas capped the quantity of sugar eligible for price support, so as to limit structural surpluses and their budgetary cost. Quotas were allocated in a top-down cascade. Member states allocated them to sugar processors, which in turn contracted with farmers and assigned them delivery rights, expressed in tonnes of beet. In practice, these rights were rolled over almost automatically from one year to the next, with no annual reallocation. Since sugar beet has been the most profitable arable crop per hectare in France over the past decades (Appendix Figure A2), farms holding delivery rights had no incentive to give them up, while the fixed overall quota left little room for new entrants.

A closed market. Table 1 documents the resulting persistence of sugar beet cultivation across farms. Among farms that did not grow sugar beet in 1988, only 2.5% grew it in 2015. For the other main arable crops, entry rates over the same period range from 25% for rapeseed to 55% for maize. The share of land devoted to sugar beet is accordingly very persistent: its correlation between 1988 and 2015 reaches 0.84, against 0.47 to 0.70 for the other crops. Because of the market structure imposed by the quota system, the allocation of sugar beet across farms in 2015 thus largely reflects decisions made three decades earlier, rather than farms' recent choices or their expectations about future regulation. Within a given locality, exposure to the ban was inherited rather than chosen.

Table 1. Persistence of Crop Cultivation across Farms, 1988–2015

	Sugar beet	Wheat	Barley	Maize	Rapeseed
P(grown in 2015 | not grown in 1988)	0.025	0.350	0.311	0.546	0.249
Correlation of land share, 1988–2015	0.835	0.699	0.467	0.550	0.509

Notes: The first row reports the share of farms growing each crop in 2015 among farms that did not grow it in 1988. The second row reports the correlation between the share of farm land devoted to each crop in 1988 and in 2015. The sample consists of farms observed in both the 1988 Agricultural Census and the 2015 Graphical Land Parcel Register.

Sample and treatment. Two further conditions make the neonicotinoid ban a suitable natural experiment. First, the shock must be economically large for exposed farms. Sugar beet accounts for a substantial share of the land of specialized farms and is the most profitable crop per hectare, so that its share in their revenue is even larger. Second, exposure to the ban requires actually using neonicotinoids. Since 96% of sugar beet farms did so in 2017, I treat all sugar beet growers as users. If some highly exposed farms did not use neonicotinoids, the estimates would understate the effect of the ban

[8] Council Regulation (EEC) No 1009/67 of 18 December 1967 on the common organisation of the market in sugar.

and should therefore be read as a lower bound.

I restrict the baseline sample to farms that grew sugar beet between 2015 and 2017, so as to compare farms with *a priori* similar production tools and infrastructure, and extend the control group to all farms as a robustness check. I focus on a balanced panel of farms observed every year from 2013 to 2022, so that the estimates are not driven by farm entry or exit. Farms whose sugar beet land share in 2015, three years before the ban, exceeds the 90th percentile (21% of cultivated land) are classified as exposed and form the treatment group, while the other farms form the control group. On average, the 1,560 exposed farms devoted 24% of their land to sugar beet in 2015, against 9% for the 14,042 control farms. The two groups are otherwise similar (Appendix Table A1). Exposed farms are slightly larger (159 ha against 149 ha), have higher crop production revenues (297 k€ against 230 k€), and have marginally higher land productivity, while their employment levels are nearly identical (1.19 against 1.18 full-time equivalent workers).

Specification. Indexing farms by i, cities by c, and years by t, I estimate the following event study:

$$\log Y_{it} = \sum_{k \neq 2017} \delta_k \, Exposed_i \times \mathbf{1}\{t = k\} + \mu_i + \lambda_{ct} + \varepsilon_{it}, \tag{1}$$

where Y_{it} is the outcome of interest, $Exposed_i$ indicates farms in the treatment group, μ_i are farm fixed effects, and λ_{ct} are city-by-year fixed effects. The coefficient δ_k measures the cumulative change in the outcome of exposed farms relative to control farms in year k, relative to 2017, the last year before the ban. To ease interpretation, the figures report δ_k divided by the difference in average pre-ban sugar beet land share between exposed and control farms, which is 15 percentage points in the baseline sample. The rescaled estimates measure the percentage change in the outcome associated with one additional percentage point of pre-ban sugar beet share, and are directly comparable with the continuous specification of Section 3.4. City-by-year fixed effects absorb local shocks, such as weather or pest pressure, so that exposed farms are compared with less exposed farms located in the same city. The main outcomes are revenue land productivity, value-added land productivity, and total revenues. All regressions are weighted by the square root of farm cultivated area. Standard errors are clustered at the farm level, the level at which treatment is assigned, which allows for arbitrary serial correlation in the outcomes of a given farm over time.

A causal interpretation of the estimates requires that, absent the ban, the outcomes of exposed and

control farms located in the same city would have followed parallel trends:

$$E[Exposed_i \times \mathbf{1}\{t = k\} \cdot \varepsilon_{it} \mid \mu_i, \lambda_{ct}] = 0 \quad \forall k.$$

If this condition holds, the pre-ban coefficients (δ_k for $k < 2017$) should be close to zero and statistically insignificant. The absence of pre-trends is a necessary condition, but it may not be sufficient. Two concerns remain: farms may have adjusted their sugar beet cultivation in anticipation of the ban, and exposure measured in a single year may capture transitory peaks that would revert to the mean even absent the ban. Section 3.4 addresses both concerns.

3.3 Main Results

Land productivity. I find that the land productivity of farms highly exposed to the ban declines relative to that of less exposed farms. Figure 3 reports the rescaled estimates of equation (1) for revenue land productivity, defined as total production excluding subsidies per hectare of cultivated land, and hereafter referred to as land productivity. There is no evidence of pre-trends. The effect is small and not statistically significant in 2018: farmers sow sugar beet around March and harvest it around October, so that most of the 2018 growing season took place before the ban came into force on September 1.[9] Land productivity then declines by 0.32% per percentage point of pre-ban sugar beet share in 2019, the first full season under the ban, and by 0.46% in 2022.[10] The estimates are precise: in 2022, the 95% confidence interval ranges from -0.62% to -0.29%. On average over 2018–2022, land productivity declines by 0.32% per percentage point of pre-ban sugar beet share. Given the 15-percentage-point difference in pre-ban sugar beet share between the two groups, this corresponds to a decline of about 5% for exposed farms relative to control farms.

Revenues, value added, and input costs. The decline in land productivity is driven by its numerator. Total revenues excluding subsidies decline relative to control farms, by 0.31% per percentage point of pre-ban sugar beet share in 2019 and 0.39% in 2022 (Figure 4a), close to the effect on land productivity. Revenue land productivity, however, ignores the cost of intermediate inputs. Since crop protection products are costly, one might expect the ban to reduce exposed farms' spending on intermediate inputs, partly offsetting the decline in revenues. Value-added land productivity, which nets out intermediate consumption, shows that this is not the case: it declines with the same timing and about three times as much, by 1.00% per percentage point of pre-ban sugar beet share on average over 2018–2022 and 1.17% in 2022 (Figure 4b). Expenses on crop protection products help explain this

[9] Since the French ban was adopted in 2016, farms could have anticipated it. Section 3.4 addresses this concern.

[10] The derogation covered the 2021 and 2022 growing seasons. As shown in Section 3.5, it did not reverse the reallocation of land away from sugar beet, so that the estimates for 2021 and 2022 still reflect the adjustments induced by the ban.

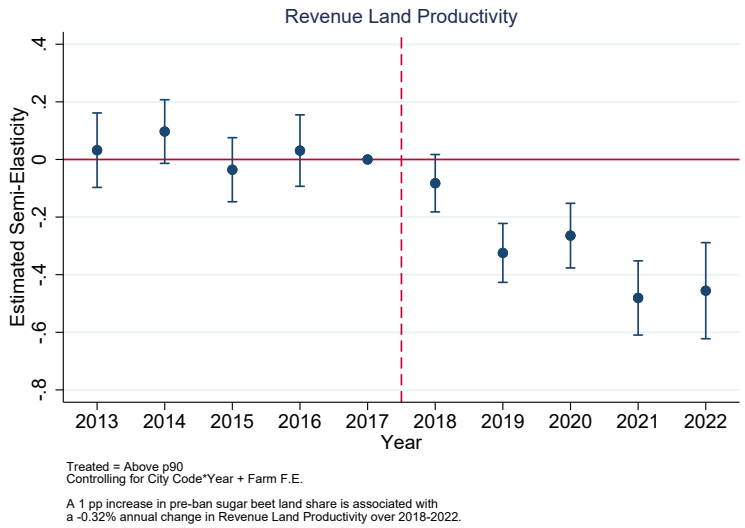


Figure 3. Effect of the Neonicotinoid Ban on Land Productivity

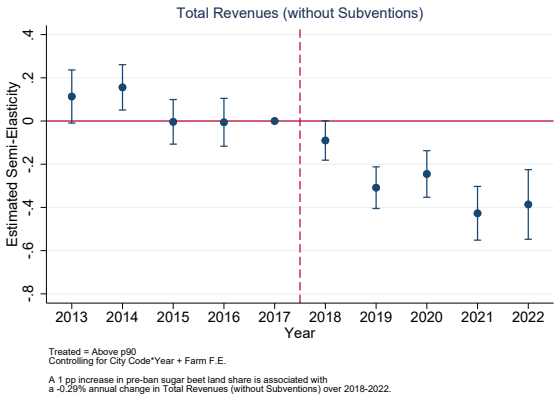
Notes: This figure reports the estimates of equation (1) for log land productivity, with 95% confidence intervals, divided by the difference in average pre-ban sugar beet land share between exposed and control farms: they measure the percentage change in land productivity per additional percentage point of pre-ban sugar beet share. Exposed farms are those above the 90th percentile of the 2015 sugar beet land share. The sample is a balanced panel, over 2013–2022, of farms that grew sugar beet in 2015–2017. The specification includes farm and city-by-year fixed effects, and standard errors are clustered by farm.

pattern (Figure 4c). Exposed farms increase their spending on crop protection products in 2019, the first full season under the ban, by 0.29% per percentage point of pre-ban sugar beet share, consistent with the replacement of neonicotinoid seed coatings by alternative insecticides, before the difference becomes statistically indistinguishable from zero. With revenues falling and input costs not declining, the loss of revenue translates more than proportionally into a loss of value added.

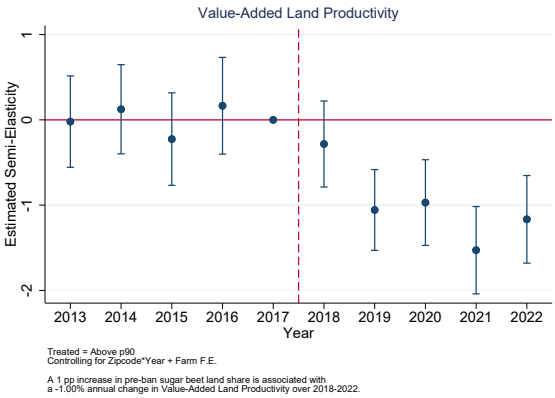
Employment and earnings. Lower activity also reduces labor demand. The wage bill of exposed farms declines relative to control farms, by 0.22% per percentage point of pre-ban sugar beet share in 2018 and 0.72% in 2022. This decline is primarily driven by lower employment, which falls by 1.15% per percentage point in 2022 (Appendix Figure A3).

3.4 Threats to Identification and Robustness

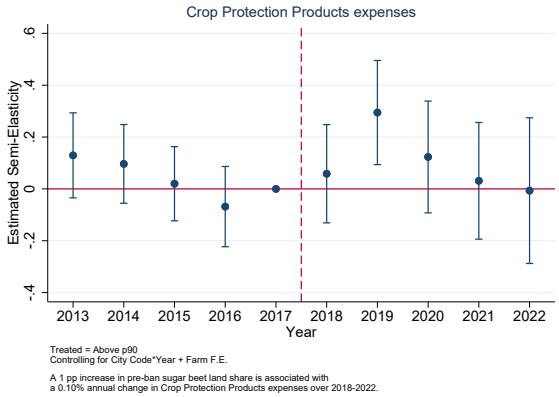
Anticipation and mean reversion. Two concerns could bias the baseline estimates. First, since the French ban was adopted in 2016, farms may have adjusted their sugar beet cultivation in anticipation of it. Second, defining exposure in a single year may select farms whose sugar beet share was temporarily high; such farms would mechanically revert toward the mean, even absent the ban. To address both concerns, I redefine exposure using the land devoted to sugar beet in 1988, as recorded in



(a) Total revenues (excluding subsidies)



(b) Value-added land productivity



(c) Expenses on crop protection products

Figure 4. Effect of the Neonicotinoid Ban on Revenues, Value Added, and Input Costs

Notes: This figure reports the estimates of equation (1), along with 95% confidence intervals, for log total revenues excluding subsidies (panel a), log value added per hectare (panel b), and log expenses on crop protection products (panel c). Estimates are rescaled as in Figure 3. The sample is the same as in Figure 3. Panels (a) and (c) include farm fixed effects and city-by-year fixed effects, and panel (b) farm fixed effects and zipcode-by-year fixed effects.

the Agricultural Census, for a subsample of about 5,000 farms. Since neonicotinoids were not yet on the market in 1988, exposure measured in that year cannot reflect any anticipation of their regulation, and the thirty-year lag purges exposure of transitory fluctuations. Figure 5 reports the results, defining exposed farms as those above the 90th percentile of the 1988 sugar beet share and using zipcode-by-year fixed effects. The results are reassuring: there is no evidence of pre-trends, and the effect is of the same order of magnitude as in the baseline specification, at 0.56% per percentage point of 1988 sugar beet share on average over 2018–2022, against 0.32% in the baseline. Given the smaller sample, the estimates are less precise: in 2022, the 95% confidence interval ranges from -0.95% to -0.07%.

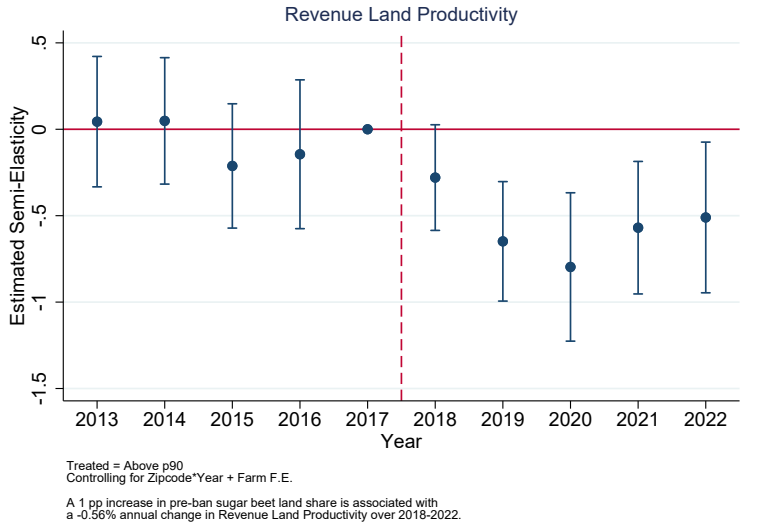


Figure 5. Effect of the Neonicotinoid Ban on Land Productivity, Exposure Defined in 1988

Notes: This figure reports the estimates of equation (1) with log land productivity as the outcome, along with 95% confidence intervals. Exposed farms are those whose sugar beet land share in 1988, measured in the Agricultural Census, exceeds the 90th percentile. Estimates are divided by the difference in average 1988 sugar beet land share between exposed and control farms. The specification includes farm fixed effects and zipcode-by-year fixed effects. Standard errors are clustered at the farm level.

Robustness. I now assess the robustness of the land productivity response. First, I replace the binary treatment with a continuous measure of exposure, the pre-ban sugar beet land share $\bar{s}_i$, expressed in percentage points:

$$\log Y_{it} = \sum_{k \neq 2017} \delta_k \, \bar{s}_i \times \mathbf{1}\{t = k\} + \mu_i + \lambda_{ct} + \varepsilon_{it}. \tag{2}$$

Figure 6 reports the estimates. Each additional percentage point of pre-ban sugar beet share reduces land productivity by 0.44% per year on average over 2018–2022, compared with 0.32% in the baseline specification. The two estimates are of the same order of magnitude. The continuous specification

exploits variation in exposure across the entire distribution of sugar beet shares, while the baseline specification compares the average exposure of two groups.

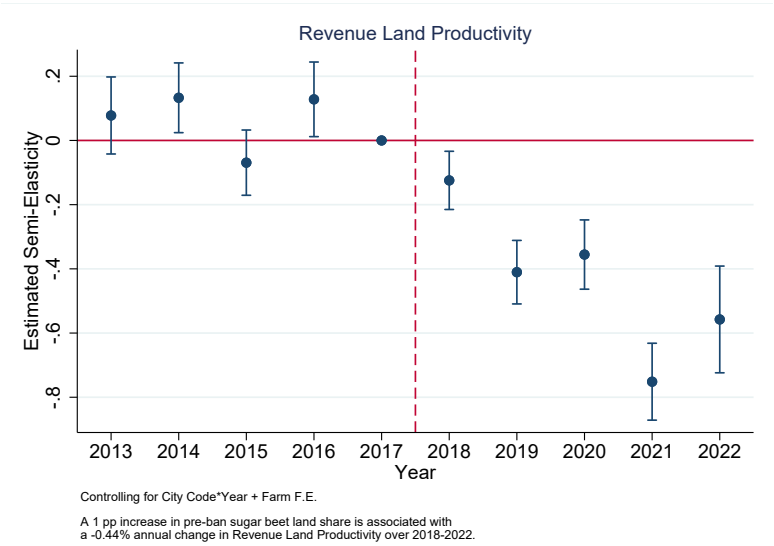


Figure 6. Effect of the Neonicotinoid Ban on Land Productivity, Continuous Exposure

Notes: This figure reports the estimates of equation (2) with log land productivity as the outcome, along with 95% confidence intervals. Exposure is the pre-ban sugar beet land share. Estimates measure the percentage change in land productivity associated with one additional percentage point of pre-ban sugar beet share. The sample and fixed effects are the same as in Figure 3.

Second, the results are similar when exposure is defined using higher thresholds of the 2015 sugar beet share (Appendix Figure A4). With the 95th percentile, land productivity declines by 0.31% per percentage point of pre-ban sugar beet share on average over 2018–2022, as in the baseline. With the 99th percentile, the decline is smaller, at 0.20%, and less precisely estimated, since only 1% of farms lie above this threshold. Third, the results are robust to replacing city-by-year fixed effects with zipcode-by-year or county-by-year fixed effects (Appendix Figure A5). The estimates are slightly larger with these coarser fixed effects, at 0.37% and 0.40% per percentage point on average over 2018–2022, respectively. Fourth, the estimates remain stable when the panel is balanced over a longer period, starting in 2010 (Appendix Figure A6). The average effect is 0.38% per percentage point, against 0.37% for the panel starting in 2013, and there is no evidence of pre-trends over 2010–2016.

Finally, the baseline control group consists of farms that also grew some sugar beet between 2015 and 2017, and were therefore partly exposed to the ban. If sugar beet production involves specific investments, these farms may themselves have borne adjustment costs, in which case the baseline specification would understate the effect of the ban. Appendix Figure A7 extends the sample to all

farms, regardless of whether they grew sugar beet before the ban, and defines exposed farms as those above the 99th percentile of the sugar beet share, which corresponds to 24% of cultivated land. The estimated effect remains of the same order of magnitude, at 0.23% per percentage point of pre-ban sugar beet share on average over 2018–2022.

EU sugar quota regime. The abolition of EU sugar quotas in September 2017, one year before the ban, could also confound the estimates, through three channels. None of them, however, is likely to drive the results. First, farms may have adjusted their sugar beet cultivation ahead of the abolition. Defining exposure in 2015, or in 1988 as above, rules out this concern. Moreover, the sugar beet land share of exposed farms shows no pre-trend (Figure 8a, discussed in Section 3.5), which rules out anticipatory adjustments. Second, the abolition could have depressed sugar beet prices, hurting the farms most specialized in sugar beet the most. Sugar beet prices, however, remained broadly stable between 2017 and 2020 and even increased sharply between 2020 and 2022 (Appendix Figure A8). The price of sugar is determined on international markets and in turn determines the price of sugar beet, which French farmers take as given. Third, by lifting the cap on production, the abolition encouraged sugar beet cultivation for all farms. Its differential effect on the two groups is ambiguous: exposed farms already have assets suited to sugar beet and may expand its production, while less exposed farms may take this opportunity to grow more of it. Appendix Figure A9 plots the average sugar beet land share of the two groups. In 2017, the first growing season without quotas, both groups expand their sugar beet acreage by similar amounts, by 1.7 and 2.2 percentage points for exposed and control farms, respectively. Their sugar beet land shares then trend downward after the ban, more steeply for exposed farms. The abolition of quotas thus does not appear to have affected the two groups differently. The next subsection decomposes the effect of the ban along three margins: sugar beet yields, relative prices, and the reallocation of land across crops.

3.5 Decomposing the Effect

Why does land productivity decline? A farm's land productivity is the sum, across crops, of the share of land devoted to each crop multiplied by its revenue per hectare (in euros per hectare), which is itself the product of yield (in tonnes per hectare) and price (in euros per tonne). The ban can therefore lower the relative land productivity of exposed farms through three channels: lower sugar beet yields, a reallocation of land away from sugar beet toward less profitable crops, and changes in the price of sugar beet relative to other crops, which weigh more on farms that grow more of it. A first-order approximation, derived step by step in Appendix B, decomposes the effect of the ban on

land productivity in year t, $\bar{\gamma}_t$, into these three channels and a residual:

$$\bar{\gamma}_t \approx \frac{1}{\bar{Y}_0}\Big[\underbrace{\tilde{s}_0\, q_0\, \delta\bar{y}_t^b}_{\text{Yield shocks}} + \underbrace{\bar{\pi}_0\, \Delta s_t^b}_{\text{Induced reallocation}} + \underbrace{\tilde{s}_0\big(\bar{y}_0^b\, \delta q_t - \delta\bar{p}_t^o\big)}_{\text{Relative price effects}} + \underbrace{R_t}_{\text{Residual}}\Big], \tag{3}$$

where $\bar{Y}_0$ is average pre-ban land productivity, $\tilde{s}_0$ is the pre-ban difference in sugar beet land share between exposed and control farms, q_0 and $\bar{y}_0^b$ are the pre-ban price and yield of sugar beet, $\delta\bar{y}_t^b$ and δq_t are their changes since 2017, and $\delta\bar{p}_t^o$ is the change in the average revenue per hectare of the other crops grown by sugar beet producers. $\bar{\pi}_0$ is the pre-ban gap in revenue per hectare between sugar beet and other crops, and Δs_t^b is the change in the sugar beet land share of exposed farms relative to control farms. The residual R_t collects the remaining terms, including second-order interactions. Appendix B details the underlying assumptions, and Appendix Table B2 summarizes the interpretation of each term.

Estimating these terms requires separating quantities, prices, and yields by crop, which balance-sheet data do not allow. I therefore rely on the subsample of farms from the main sample that are also surveyed in the FADN, which records the area, quantity, and value of production of each crop. Figure 7 reports the results, averaged over 2018–2022. As for the event-study estimates, each term is divided by the difference in average pre-ban sugar beet land share between exposed and control farms, so that the total effect corresponds to the baseline estimate of 0.32% per percentage point of pre-ban sugar beet share.

Yield shocks. I first measure the direct effect of the regulation on the productivity of the product line it targets, here sugar beet. I compute the change in average sugar beet yields after the ban and combine it with the pre-ban sugar beet price and the difference in sugar beet land share between the two groups. Because exposed farms initially devote more land to sugar beet, the same decline in yields, due in particular to beet yellows virus, weighs more on their land productivity. Lower sugar beet yields reduce land productivity by 0.12% per percentage point of pre-ban sugar beet share, which represents only 37% of the total effect of 0.32%.

Induced reallocation. A regulatory constraint on one product line can also lead firms to reorganize production across their product lines. Lower and riskier sugar beet yields give exposed farms an incentive to shift land away from sugar beet. This reallocation is costly, since sugar beet remains the most profitable arable crop per hectare (Appendix Figure A2). Farms may nonetheless reallocate to limit the risk of large losses in the event of a severe epidemic, against which they no longer have an effective means of control, and to diversify their crop rotation. Consistent with this mechanism, the

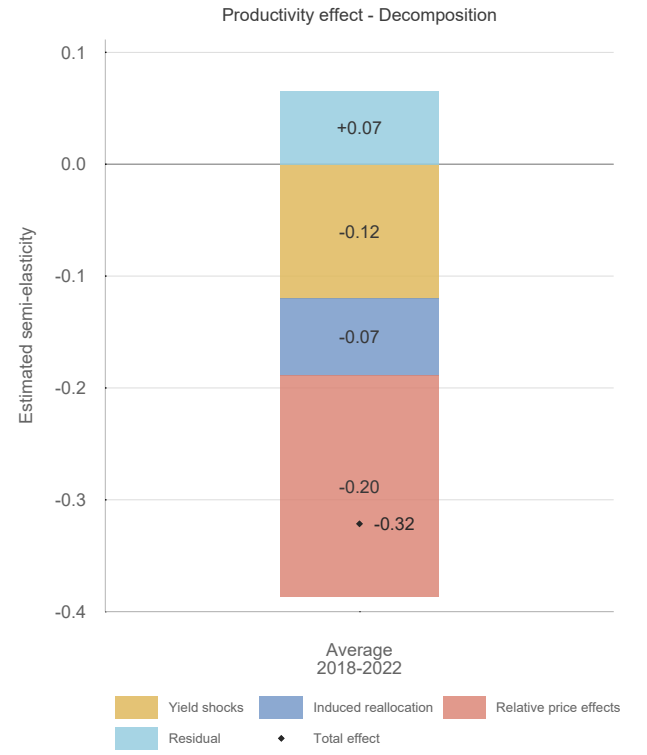


Figure 7. Decomposition of the Effect of the Neonicotinoid Ban on Land Productivity

Notes: This figure decomposes the average effect of the neonicotinoid ban on land productivity over 2018–2022 into the terms of equation (3). Each term is divided by the difference in average pre-ban sugar beet land share between exposed and control farms, so that it measures the percentage change in land productivity associated with one additional percentage point of pre-ban sugar beet share. The total effect corresponds to the baseline estimates of Figure 3. The terms of the decomposition are computed on the farms from the main sample that are also surveyed in the FADN.

sugar beet land share of exposed farms declines relative to that of control farms after the ban, with no pre-trend (Figure 8a). In 2022, it is 0.21 percentage point lower per percentage point of pre-ban sugar beet share, that is, about 3 percentage points lower given the 15-percentage-point gap between the two groups. Notably, the derogations granted for the 2021 and 2022 growing seasons did not alter the trajectory of the sugar beet land share. This suggests that regulatory changes induce a reorganization of the mix of product lines within the production function, which requires structural adjustments. A short-term derogation, without long-term visibility, is not sufficient to reverse such organizational choices. In terms of policy, this implies that regulation must be predictable and credible to shape firms' organizational choices.

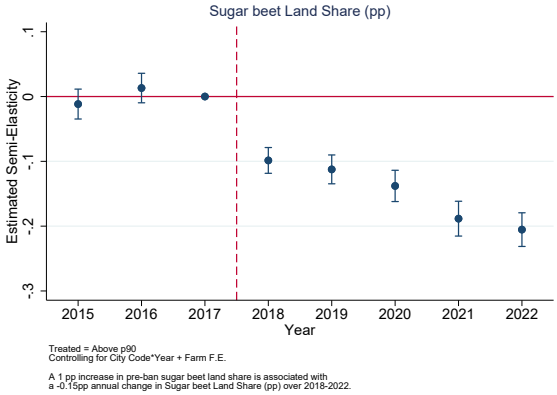
This reallocation also operates through crop rotations. Some crops, cereals in particular, rotate well with sugar beet, while others, such as rapeseed, rotate poorly with it, whether grown before or after.[11] By reducing the incentive to grow sugar beet, the ban relaxes a constraint on rapeseed cultivation. Figure 8b confirms that the rapeseed land share of exposed farms increases after the ban, by 0.08 percentage point per percentage point of pre-ban sugar beet share in 2022, that is, by about 1 percentage point.

More generally, the ban reshapes the entire crop mix of exposed farms, beyond sugar beet and rapeseed. To capture this, I compute a dissimilarity index that measures how much the crop mix of a farm in a given year differs from its crop mix in 2017.[12] Figure 8c shows that the two groups follow a common trend before the ban. Afterwards, the crop mix of exposed farms gradually departs more markedly from its pre-ban composition. The difference becomes statistically significant in 2021 and, in 2022, the dissimilarity index of exposed farms is 0.08 percentage point higher per percentage point of pre-ban sugar beet share, that is, about 1.2 percentage points higher than that of control farms. The ban thus induces a broader reorganization of the production function of exposed farms. Overall, induced reallocation reduces land productivity by 0.07% per percentage point of pre-ban sugar beet share, or 21% of the total effect.

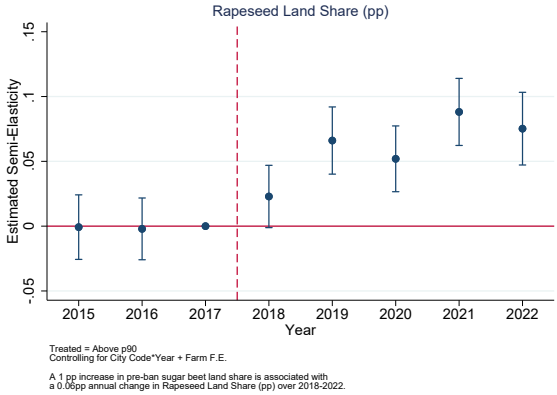
Relative price effects. Finally, sugar beet prices remained broadly stable between 2017 and 2020 and increased sharply between 2020 and 2022 (Appendix Figure A8). The relative price effect therefore reflects the faster increase in the prices of the other crops typically grown by sugar beet producers,

[11] Jeangros and Courvoisier (2019) and Fenz et al. (2023) detail crop rotation guidelines in the context of European agriculture. Rapeseed and sugar beet are both hosts of the beet cyst nematode, so that growing them in close succession increases pest pressure on both crops.

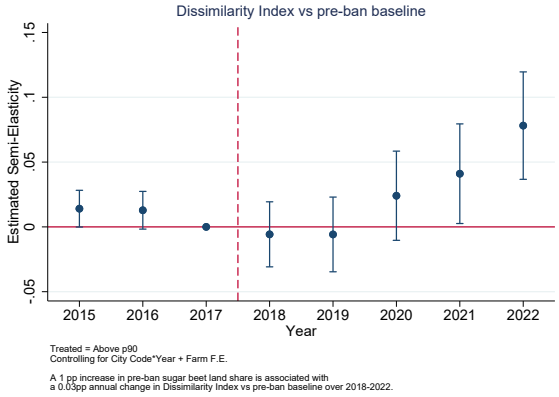
[12] For each farm i, the dissimilarity index is defined as $D_{it} \equiv \frac{1}{2}\sum_c |s_{ict} - s_{ic,2017}|$, where c indexes crops and s_{ict} is the share of the land of farm i devoted to crop c in year t. The index ranges from 0, when the crop mix is unchanged, to 1, when the farm grows none of the crops it grew in 2017. It is the crop-level counterpart of the occupational dissimilarity index used by Aghion et al. (2024).



(a) Sugar beet land share



(b) Rapeseed land share



(c) Crop mix dissimilarity index

Figure 8. Effect of the Neonicotinoid Ban on Crop Allocation

Notes: This figure reports the estimates of equation (1), along with 95% confidence intervals, with the share of farm land devoted to sugar beet (panel a), the share devoted to rapeseed (panel b), and the dissimilarity index between the crop mix of the farm and its crop mix in 2017 (panel c) as outcomes, in level. Estimates are divided by the difference in average pre-ban sugar beet land share between exposed and control farms, so that they measure the change in the outcome, in percentage points, associated with one additional percentage point of pre-ban sugar beet share. Land shares are measured in the Graphical Land Parcel Register, available from 2015. The sample and fixed effects are the same as in Figure 3.

in particular the surge in cereal and oilseed prices over the period. Because exposed farms devote a larger share of their land to sugar beet, the resulting decline in the relative price of sugar beet lowers their land productivity more. Relative price effects reduce land productivity by 0.20% per percentage point of pre-ban sugar beet share, or 62% of the total effect, while the residual term is positive (+0.07%). Unlike the first two terms, this component does not reflect an effect of the ban: it is mainly driven by international commodity markets.

Appendix Figure A10 shows how the contribution of each term evolves over time. The contribution of induced reallocation grows steadily, from 0.04% per percentage point of pre-ban sugar beet share in 2018 to 0.10% in 2022, consistent with a gradual and persistent reorganization of the production function. The contribution of sugar beet yields follows the evolution of pest pressure. The contribution of relative prices is the most volatile, as it follows world agricultural commodity markets.

Overall, the direct effect of the ban, through lower productivity of the targeted good (sugar beet yields), accounts for 37% of the total effect, and its indirect effect, through the induced reorganization of production across product lines (sugar beet land share), for 21%. Together, these two channels account for only slightly more than half of the total effect. The remainder stems from relative price movements, which are driven by world markets and orthogonal to regulatory policy. This distinction matters for policy: a scheme compensating farmers for the productivity losses caused by the regulation should be calibrated on the first two, structural, channels only, since the third is cyclical.

3.6 Environmental Effects of the Ban

The ban was motivated by the harmful effects of neonicotinoids on pollinators and aquatic insects (European Food Safety Authority, 2018). So far, I have studied its consequences for the farms that relied on these insecticides, but not its effects on this primary objective. In this subsection, I examine whether the ban, by protecting pollinators, raised the productivity of neighboring farms. I find no effect on neighboring farms on average, but a positive effect on the land productivity of orchards, whose yields depend heavily on pollination. By contrast, the ban has no effect on vineyards, which do not rely on pollinators.

Mechanism. Figure 9 summarizes the mechanism. Before the ban, the neonicotinoid seed coatings used in sugar beet cultivation harm local pollinator populations.[13] Fewer pollinators reduce the

[13] Although sugar beet is harvested before it flowers, neonicotinoids applied as seed coatings are systemic and persistent. They spread to soils, water, field margins, and succeeding crops, through which pollinators can be exposed (European Food Safety Authority, 2018).

pollination, and therefore the yields, of neighboring crops that depend on pollinators. The ban shuts down this channel: as pollinator populations recover, pollination is restored and the yields of neighboring crops increase. If this mechanism is at play, the ban should generate positive environmental spillovers, raising the productivity of farms located near sugar beet fields.

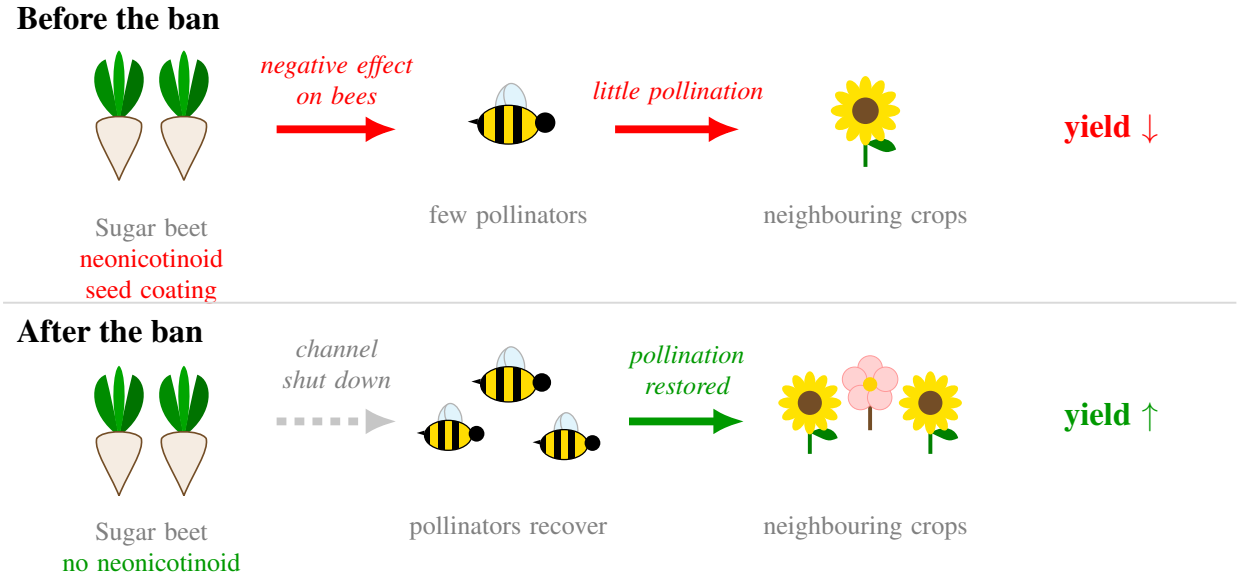


Figure 9. Environmental Spillovers of the Neonicotinoid Ban: Mechanism

Measuring local exposure. Since the Graphical Land Parcel Register records the crop grown on every parcel each year, I can measure the exposure of each parcel in France to the neonicotinoids used on neighboring fields. For each parcel i, exposure to sugar beet cultivation is the area of the sugar beet parcels located within a radius R, weighted by a decreasing function of their distance:

$$E_i^{\text{beet}} = \sum_{j \in \mathcal{B},\, d_{ij} \leq R} a_j \, e^{-d_{ij}/\rho}, \tag{4}$$

where $\mathcal{B}$ is the set of parcels cultivated with sugar beet, a_j is the area of parcel j, and d_{ij} is the distance between the centroids of parcels i and j. This exponential kernel follows the standard specification of pollination service models, in which resources are discounted with distance by $e^{-d/\alpha}$, where α is the typical foraging distance of pollinators (Lonsdorf et al., 2009). I set the decay parameter to $\rho = 600$ metres and the cutoff to $R = 1{,}500$ metres. These values are of the same order as the distances from natural habitat at which pollinator visitation rates and pollinator species richness fall to half their maximum in a meta-analysis of 23 studies, 0.6 and 1.5 kilometres respectively (Ricketts et al., 2008), and lie within the range of typical foraging distances of bees (Greenleaf et al., 2007). Appendix Figure A11 illustrates the construction. Since post-ban changes in crop allocation are themselves an

outcome of the ban, I fix exposure at its pre-ban average over 2015–2017. Farm-level exposure is the area-weighted mean of the exposure of the farm's parcels.

Figure 10 maps exposure across France. It follows the geography of sugar beet cultivation, but also varies substantially within local areas. Appendix Figure A12 illustrates this variation with two fictitious farms located in the same zipcode. Because more sugar beet is grown near its fields, the second farm is more exposed to neonicotinoids than the first, and should therefore benefit more from the ban, if at all.

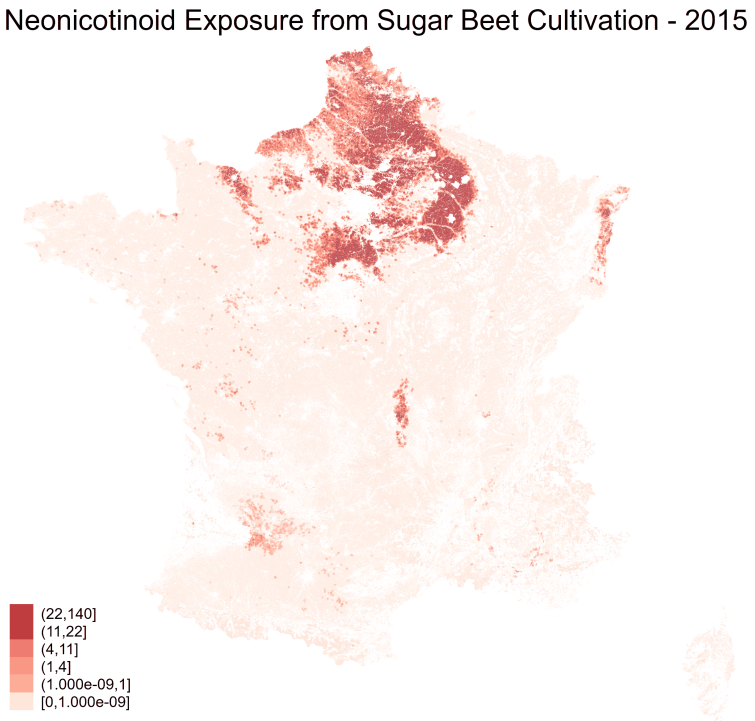


Figure 10. Local Exposure to Neonicotinoids from Sugar Beet Cultivation in 2015

Notes: This map reports, for each land parcel, its exposure to sugar beet cultivation in 2015, as defined in equation (4). Source: Graphical Land Parcel Register.

Exposure through other crops. The ban also applied to the neonicotinoid seed coatings used on wheat and barley. Growers of these crops were, however, much less dependent on neonicotinoids than sugar beet growers: alternative crop protection products were effective, and the ban had a negligible effect on their productivity[14]. This distinction matters for environmental spillovers, since pollinators may be exposed to the neonicotinoids used on any of the three crops. I therefore construct three exposure measures, E_i^{beet}, E_i^{wheat}, and E_i^{barley}, each defined as in equation (4).

[14]This partly explains why the derogation was limited to sugar beet.

Specification. Indexing farms by i, zipcodes by z, and years by t, I estimate the following event study:

$$\log Y_{it} = \sum_{c \in \mathcal{C}} \sum_{k \neq 2017} \beta_k^c \log E_i^c \times \mathbf{1}\{t = k\} + \alpha_i + \lambda_{zt} + \varepsilon_{it}, \qquad \mathcal{C} = \{\text{beet}, \text{wheat}, \text{barley}\}, \quad (5)$$

where Y_{it} is the land productivity of farm i in year t, E_i^c is its pre-ban exposure to crop c, α_i are farm fixed effects, and λ_{zt} are zipcode-by-year fixed effects. The coefficients β_k^{beet} are elasticities: they measure the change in land productivity, relative to 2017, associated with a 1% higher pre-ban exposure to sugar beet cultivation, holding exposure to wheat and barley constant. Zipcode-by-year fixed effects absorb local shocks, so that the estimates compare neighboring farms that differ in their exposure, as in the example of Appendix Figure A12. To isolate the externality from the direct effect of the ban documented above, the sample excludes farms that grow sugar beet themselves.

All neighboring farms. Figure 11 reports the estimates of β_k^{beet} for all farms. I find no evidence of positive spillovers on average: the estimated elasticities remain close to zero after the ban. This average null effect may, however, conceal different realities. The ban may indeed have no effect on neighboring farms. Alternatively, neighboring farms may have adapted to the presence of neonicotinoids before the ban, for instance by growing crops that depend less on pollination when located near intensive sugar beet growers, which would mute the effect of the ban on their productivity.

Orchards. To distinguish between these interpretations, I focus on a specific category of farms: those with at least one orchard parcel. Orchards are well suited to this test for two reasons. First, fruit trees depend heavily on pollination, so that their productivity should respond directly to changes in pollinator populations. Second, fruit trees are perennial and are not part of annual crop rotations: once planted, they cannot be relocated, whatever the use of neonicotinoids by neighboring farms. Their location is therefore unlikely to reflect an adaptation to neighboring exposure. Panel (a) of Figure 12 reports the estimates of equation (5) for orchard farms. The land productivity of more and less exposed orchard farms follows a common trend before the ban. After the ban, the land productivity of the orchard farms that were initially more exposed increases more. The effect builds up gradually and becomes statistically significant in 2022, when it reaches an elasticity of 0.034: a 10% higher pre-ban exposure to sugar beet cultivation is associated with a 0.34% higher land productivity. For a category of crops that depends heavily on pollinators, the ban thus generates positive spillovers on productivity. These estimates are, if anything, conservative: the derogation re-authorized neonicotinoid seed coatings on sugar beet for the 2021 and 2022 growing seasons, so that farms located near sugar beet fields were again exposed to these insecticides in these two years.

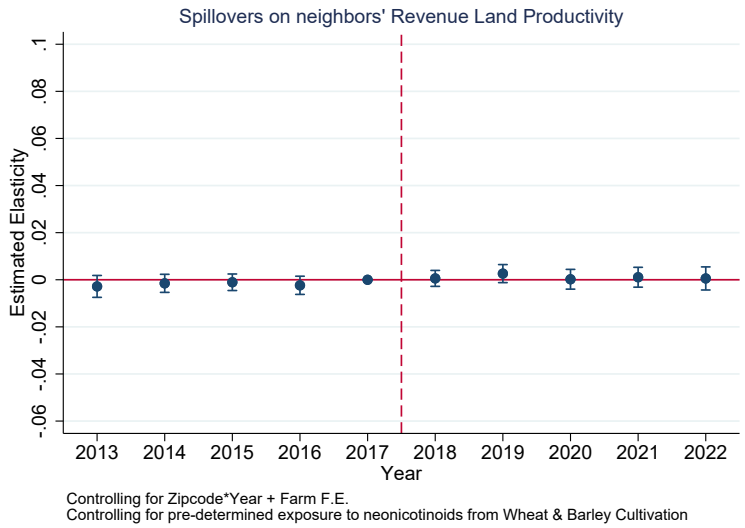


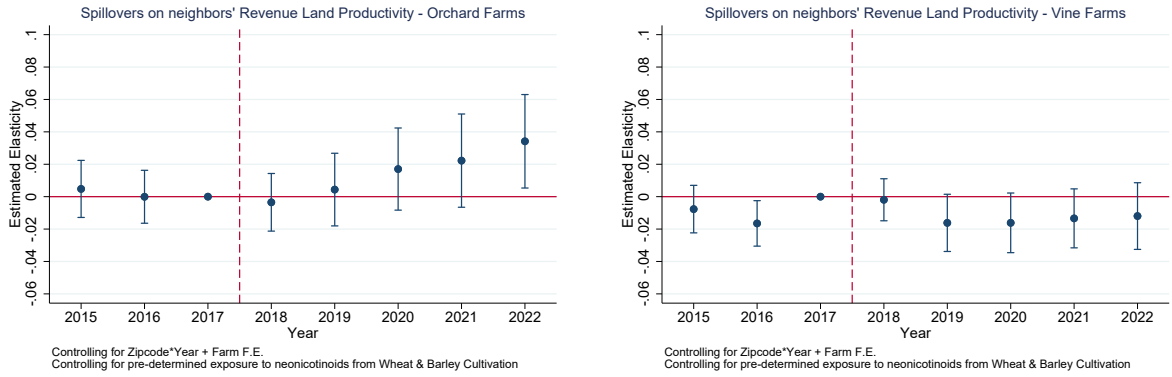
Figure 11. Spillover Effects of the Neonicotinoid Ban on Neighboring Farms

Notes: This figure reports the estimates of β_k^{beet} in equation (5) with log land productivity as the outcome, along with 95% confidence intervals. The specification controls for pre-ban exposure to wheat and barley cultivation, interacted with year indicators, and includes farm fixed effects and zipcode-by-year fixed effects. The sample consists of farms that do not grow sugar beet, observed from 2013 to 2022.

Robustness. I conduct two robustness checks, reported in Appendix Figure A13. First, the response of orchard farms may depend on how specialized they are in fruit production. Controlling for their initial orchard land share, interacted with year indicators, leaves the trajectory of the estimates qualitatively unchanged, although the effect is smaller and less precisely estimated, with an elasticity of 0.023 in 2022 that is not statistically significant at the 5% level (panel a). Second, the spillover may stem not from neonicotinoids, but from a change in the crops grown nearby. Sugar beet is harvested before it flowers and therefore does not attract bees. By contrast, rapeseed, which tends to replace sugar beet after the ban (Figure 8b), flowers before harvest. The partial replacement of a bee-neutral crop by a bee-friendly one could thus attract more pollinators and benefit neighboring orchards, independently of the ban on neonicotinoids. To test this channel, I add contemporaneous exposure to rapeseed cultivation, constructed as above, as a control. This variable is a bad control, since it is itself affected by the ban, but it helps document the channel. The results remain qualitatively similar, with an elasticity of 0.035 in 2022, at the margin of statistical significance (panel b), suggesting that the spillover operates through the effect of neonicotinoids on pollinators rather than through the shift toward bee-friendly crops.

Falsification test: vineyards. Finally, I run a falsification test on vineyards. Like fruit trees, vines are fixed and do not rotate, so that vineyards face the same constraints on their location as orchards.

Unlike fruit trees, however, vines do not require external pollinators such as bees, since they are mostly self-pollinating. If the spillover operates through pollinators, the ban should therefore have no effect on vineyards. Panel (b) of Figure 12 confirms this prediction: the ban has no positive effect on the land productivity of vineyards, whose estimates remain close to zero, and if anything slightly negative, both before and after the ban. This supports the view that the externality on neighboring farms operates through pollinator populations.



(a) Orchards

(b) Vineyards

Figure 12. Spillover Effects of the Neonicotinoid Ban on Orchards and Vineyards

Notes: This figure reports the estimates of β_k^{beet} in equation (5) with log land productivity as the outcome, along with 95% confidence intervals. The sample consists of farms with at least one orchard parcel (panel a) or vineyard parcel (panel b), observed from 2015 to 2022. The specification controls for pre-ban exposure to wheat and barley cultivation, interacted with year indicators, and includes farm fixed effects and zipcode-by-year fixed effects.

Overall, these results suggest that the ban raised the productivity of a specific category of neighboring farms, those that rely heavily on pollinators. Two mechanisms could explain this effect: existing pollinator populations may have grown after the ban, or beekeepers may have seen the ban as an opportunity to install more hives (Frank et al., 2026). In either case, the results are consistent with a gradual recovery of honey bee populations after the ban, which in turn raised the productivity of neighboring farms that rely on pollinators.

4 From One Ban to Many: How Alternatives Shape the Cost of Bans

The neonicotinoid ban is an illustrative but specific case: it concerned a single crop, and no effective substitute was available on the market. In this section, I extend the analysis to all bans on crop protection products that affected French field crops in 2017 and 2018. On average, these bans reduce the land productivity of exposed farms by about 3% after five years. This average conceals large

differences across bans. When the banned products have no substitute, land productivity declines by about 10%, a pattern close to that of the neonicotinoid ban. When substitutes are scarce, the decline is halved, and when they are abundant, the bans have no effect. I first present the empirical approach (Section 4.1) and then the results (Section 4.2).

4.1 Empirical Approach

Measuring exposure to pesticide bans. No administrative source records which crop protection products each French farm uses on each crop. The neonicotinoid case could bypass this limitation, since virtually all sugar beet growers used these insecticides. In general, however, exposure must be measured. I rely on the 2017 Field Crop Cultural Practices Survey, which records, for about 20,000 parcels, the crop grown, every crop protection product applied, and the identifier of the farm. Matching the survey with the Graphical Land Parcel Register measures the weight of the surveyed crop in each farm. I restrict the analysis to field crops, excluding horticulture, viticulture, and greenhouse farming, so as to work with farms whose land use is relatively homogeneous.

The information provided by the survey is only economically relevant if the surveyed crop matters for the farm. I therefore keep farms that cultivated this crop intensively in 2015, two years before the survey: its share in the farm's cultivated land must exceed both the average share among French farms that grow it and 10%.[15] Using the E-Phy database, which records the marketing authorizations of all crop protection products, I classify a farm as exposed if, in 2017, it used on the surveyed crop a product whose use on this crop was banned in 2017 or 2018. The final sample consists of 863 exposed farms and 2,304 farms that were not exposed to any ban on an economically relevant crop.

Exposed farms are slightly larger than non-exposed farms, with 170 ha of cultivated land on average against 153 ha (Appendix Table A2). They also have a larger stock of machines (168 k€ against 146 k€) and slightly higher land productivity (2.19 k€ per hectare against 2.04 k€). As in the neonicotinoid case, exposed farms are thus slightly larger and slightly more productive than other farms (Section 3.2).

Specification. I estimate an event study similar to that of Section 3.2. Indexing farms by i,

[15] The first condition selects farms that are specialized in the surveyed crop relative to other growers. The second excludes crops that remain marginal for the farm even when it is specialized relative to other growers, as is the case for crops that all farms grow on small areas. This approach assumes that a farm applies the same crop protection products to all its fields of a given crop.

counties by d, and years by t:

$$\log Y_{it} = \sum_{k \neq 2016} \delta_k \, ExposedBan_i \times \mathbf{1}\{t = k\} + \mu_i + \lambda_{dt} + \varepsilon_{it}, \tag{6}$$

where Y_{it} is the land productivity of farm i in year t, $ExposedBan_i$ indicates farms exposed to a ban, μ_i are farm fixed effects, and λ_{dt} are county-by-year fixed effects. The coefficient δ_k measures the cumulative change in the land productivity of exposed farms relative to non-exposed farms in year k, relative to 2016, the last year before the bans.

Identification. A causal interpretation of the estimates requires that, absent the bans, the land productivity of exposed and non-exposed farms located in the same county would have followed parallel trends:

$$E[ExposedBan_i \times \mathbf{1}\{t = k\} \cdot \varepsilon_{it} \mid \mu_i, \lambda_{dt}] = 0 \quad \forall k.$$

If this condition holds, the pre-ban coefficients should be close to zero and statistically insignificant. The absence of pre-trends is necessary but not sufficient: supply or demand shocks may coincide with a ban. The main concern is that bans may themselves be triggered by innovation. The arrival of a new active substance or of a resistant variety could both raise the productivity of farms and make regulators more willing to withdraw an older, more hazardous product. Two arguments suggest that this concern is limited. First, the timing of bans is largely set by regulatory procedures: active substances are approved for a limited period, and their renewal is assessed on hazard and risk criteria rather than on market conditions.[16] Second, if bans were triggered by the arrival of better alternatives, exposed farms would have adopted these alternatives even without the ban, and their productivity would have increased relative to that of non-exposed farms. This would bias the estimates upward, so that the negative effects documented below would understate the losses caused by the bans. Conversely, a ban coinciding with a loss of efficacy of the banned product, for instance due to pest resistance, would bias the estimates downward. Such a loss of efficacy would, however, build up gradually and generate pre-trends, which are not observed.

4.2 Results

Land productivity. Pesticide bans reduce the land productivity of exposed farms (Figure 13). There is no evidence of pre-trends. The effect builds up gradually and becomes statistically significant five years after the last pre-ban year, with a semi-elasticity of -0.03 in 2021 and 2022: the land

[16] Under Regulation (EC) No 1107/2009, active substances are approved at the EU level for up to ten years for a first approval and up to fifteen years for a renewal.

productivity of exposed farms declines by 3% relative to that of non-exposed farms. The estimates are imprecise, and only marginally significant at the 95% level. The result holds when the condition that the surveyed crop represents at least 10% of the farm's cultivated land is relaxed (Appendix Figure A14). It is consistent with the findings of Section 3, and generalizes them beyond a single product and a single crop. To the best of my knowledge, this is the first evidence, based on a systematic approach and a large sample of farms, that bans on crop protection products reduce farm productivity.

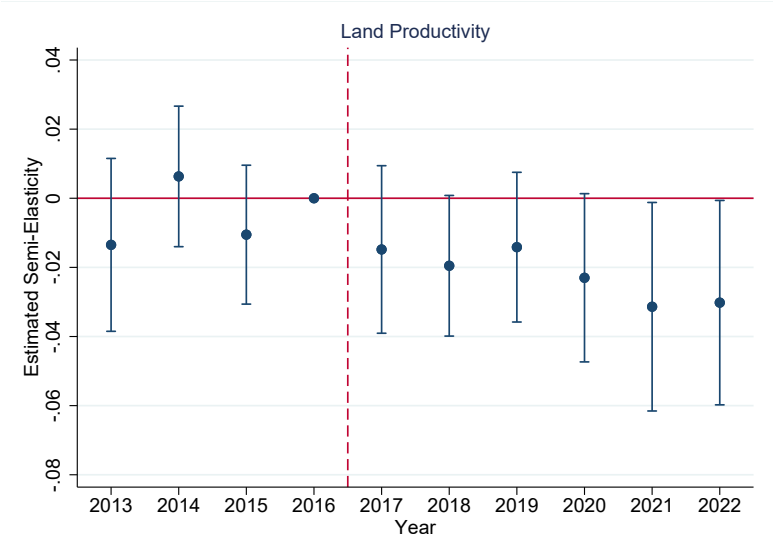


Figure 13. Effect of Pesticide Bans on Land Productivity

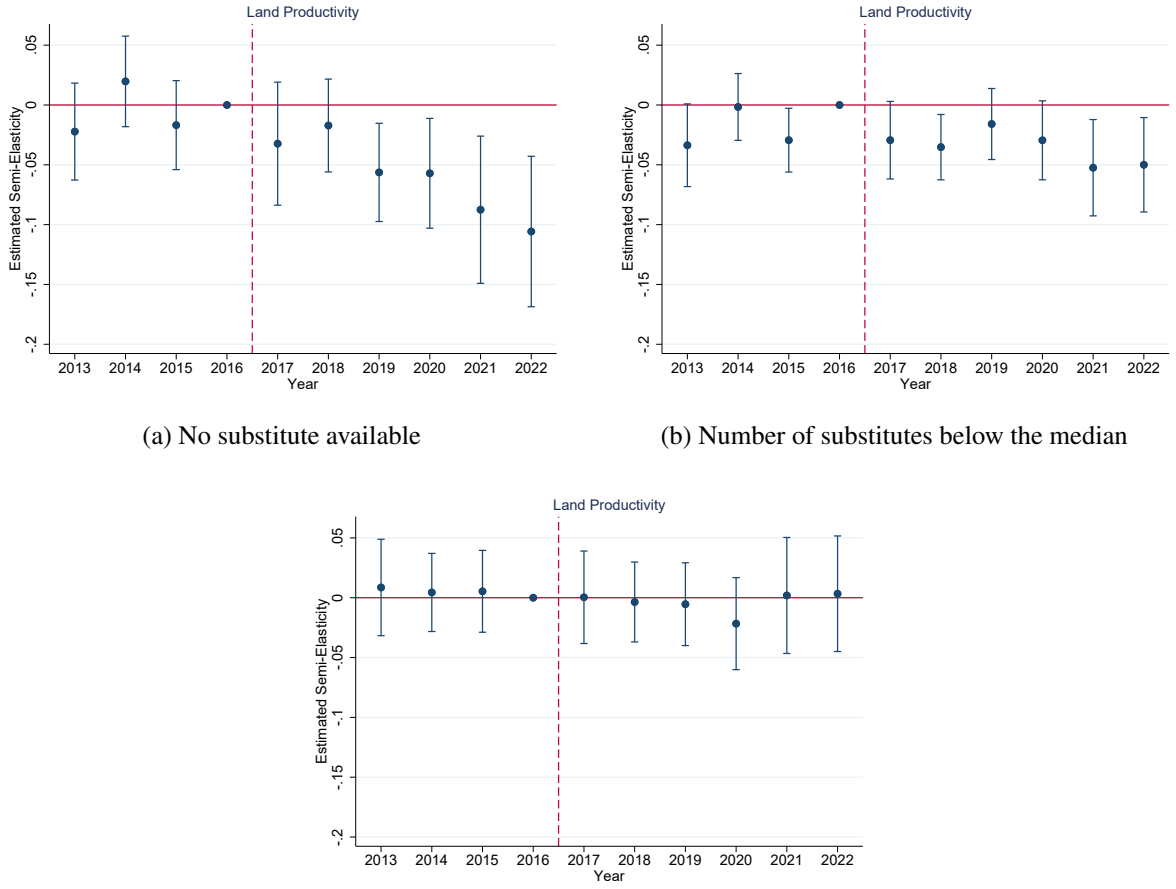
Notes: This figure reports the estimates of equation (6) for log land productivity, with 95% confidence intervals. Exposed farms are those that used, in 2017, a product whose use on an economically relevant crop was banned in 2017 or 2018. The control group consists of farms not exposed to any ban on an economically relevant crop. The specification includes farm and county-by-year fixed effects.

Heterogeneity by availability of substitutes. The average effect masks substantial heterogeneity, which depends on whether farms can replace the banned products. For each crop protection product, the E-Phy database describes its authorized uses along two dimensions: the crop on which it can be applied (e.g., wheat or barley) and its target (e.g., insects or fungi). I use this information to count, each year, the number of products authorized for each use, and therefore the number of potential substitutes for each banned product. I then split exposed farms into three groups, keeping the same control group: farms whose banned products had no substitute (139 farms), between 1 and the median number of 52 substitutes (387 farms), and more than 52 substitutes (266 farms).[17]

Figure 14 reports the estimates of equation (6) for each group. When no substitute is available, land productivity declines from 2019 onwards, and the effect grows over time to reach a semi-

[17]Exposed farms whose banned products fall into different groups are not included in this analysis.

elasticity of -0.11 in 2022 (panel a). This pattern and its magnitude are close to those of the neonicotinoid ban, under which exposed farms lost about 7% of their land productivity relative to control farms in 2022 (Section 3.3). The similarity is expected, since the neonicotinoid ban is a typical case of a ban on a product without substitute. When substitutes exist but are scarce, land productivity also declines, but by half as much, with a semi-elasticity of -0.05 in 2022 (panel b). When substitutes are abundant, the effect is a precisely estimated zero (panel c). In all three groups, there is no evidence of pre-trends.



(a) No substitute available

(b) Number of substitutes below the median

(c) Number of substitutes above the median

Figure 14. Effect of Pesticide Bans on Land Productivity, by Number of Substitutes

Notes: This figure reports the estimates of equation (6) for log land productivity, with 95% confidence intervals, for farms whose banned products had no substitute (panel a), a number of substitutes below the median of 52 (panel b), or above the median (panel c). The control group consists of farms not exposed to any ban on an economically relevant crop. The specification includes farm and county-by-year fixed effects.

Robustness. The heterogeneity by availability of substitutes is robust to alternative measures of land productivity. First, I use as denominator the cultivated land declared under the Common Agricul-

tural Policy instead of the land reported in balance-sheet data (Appendix Figure A15). This measure would be preferable, but it is only available from 2015, which limits the analysis of pre-trends. The estimates are very similar to the baseline. Second, I use crop production revenues as numerator instead of total revenues excluding subsidies (Appendix Figure A16). The pattern is again similar, with slightly larger declines for farms with no or few substitutes, and a slightly positive but noisy effect for farms with many substitutes.

Mitigation practices. The 2017 survey also records whether farms had adopted practices to reduce their use of crop protection products, and why. Appendix C uses this information to compare exposed farms that had adopted such practices for economic reasons with other exposed farms. When the banned products had no substitute, these practices do not soften the effect of the ban. When substitutes exist but are scarce, they more than offset it: in 2022, the land productivity of exposed farms with mitigation practices is about 12% higher than that of control farms, while it is about 7% lower for exposed farms without them. When substitutes are abundant, the ban has no effect in either group.

Implications for regulation. While intuitive, these results have direct implications for regulatory design. For products that pose comparable risks to health, the environment, or biodiversity, bans are much less costly for farms when substitutes are available. So far, the results therefore support the principle of not banning a product without substitutes. The losses caused by bans are concentrated on products without substitutes, and when substitutes are scarce, farms that had adopted mitigation practices avoid these losses. This view, however, is static: it takes the set of available technologies as given. Yet bans may also spur the development of new substitutes. The next section examines how accounting for innovation changes these conclusions.

5 Creating Alternatives: How Pesticide Bans Induce Innovation

Section 4 shows that bans are much less costly for farms when the banned products have substitutes, which supports the principle of not banning a product without substitutes. This conclusion, however, takes the set of available technologies as given. Accounting for innovation may overturn it. In endogenous growth models, innovation is driven by the prospect of monopoly rents (Romer, 1990; Aghion and Howitt, 1992). A ban that leaves a use without any authorized product creates a void: a firm that fills it with a new product faces little or no competition and can earn large rents. By contrast, when the banned product has many close substitutes, the market remains competitive and offers weaker incentives to innovate (Aghion et al., 2005). Bans on products without substitutes may therefore be the ones that induce the most innovation. In this section, I show that this is the case. Bans that leave a use

without substitutes increase the probability that a new active substance is authorized for this use by about 5 percentage points six to eight years later, through both the adaptation of existing substances and the introduction of new molecules.

Related literature. These questions relate to a broad literature on the effect of environmental regulation on innovation. Porter and van der Linde (1995) argue that well-designed environmental regulation can trigger innovations that partly or fully offset the costs of complying with it. The literature on directed technical change formalizes how policy can redirect innovation: Acemoglu et al. (2012) show that carbon taxes and research subsidies can redirect innovation from dirty to clean technologies, and Aghion et al. (2016) find that higher fuel prices redirect patenting in the automotive industry toward clean technologies. More generally, innovation responds to the expected size of the market. Acemoglu and Linn (2004) show that drug categories whose potential market grows because of demographic change attract more new drugs, and Finkelstein (2004) shows that health policies that raised the expected returns to vaccines against specific diseases increased clinical trials for new vaccines against these diseases. Bans can induce innovation as well: Dugoua (2025) shows that the Montreal Protocol, which scheduled the phase-out of chlorofluorocarbons (CFCs) in 1987, sharply increased patents and scientific articles on their substitutes. These substitutes were mostly new to the industry rather than new to the world: they were known compounds that had to be adapted to industrial applications. Regulation can, however, also discourage innovation when it raises the cost of firm growth (Aghion et al., 2023).

Market structure. The relevant market is no longer that of farms, which are the final users of crop protection products, but the upstream market of the agrochemical companies that develop them. This market is highly concentrated. Following the mergers of Dow and DuPont, of ChemChina and Syngenta, and of Bayer and Monsanto between 2017 and 2018, four firms (Bayer, Corteva, Syngenta Group, and BASF) account for the bulk of global sales of crop protection products (Clapp, 2021). Innovation in this market is costly and slow: developing a new active substance took about eleven years and 286 million dollars on average in the early 2010s (Phillips McDougall, 2016). Once authorized, a new substance is protected by patents and regulatory barriers to entry, so that the firms that develop it can earn substantial rents, especially on uses where it faces no competing product.

Approach. I study innovation at the level of uses, defined as crop–target pairs (e.g., aphid control in sugar beet), which correspond to the markets in which crop protection products compete. I ask whether uses that lose all their authorized products after a ban attract more innovation than uses that keep at least one alternative.

5.1 Empirical Approach

Data and regulatory events. I use the E-Phy database, maintained by the French Agency for Food, Environmental and Occupational Health & Safety (ANSES), which records all authorized crop protection products and their uses. I build a balanced panel of crop–target uses observed annually from 1960 to 2023, which records the active substances authorized each year for each use. For each use u and year t, I measure the intensity of bans as the share of the products available at the start of the year that are withdrawn during the year:

$$R_{u,t} = \frac{\text{Number of products banned in year } t}{\text{Number of products available at the start of year } t}. \tag{7}$$

The regulatory event of use u occurs in the year τ_u in which this ratio is largest, and R_{u,τ_u} measures its intensity. It ranges from 0, when no product is banned, to 1, when all the products authorized for the use are banned in the same year. A substantial number of uses experience such complete withdrawals (Appendix Figure A17). For instance, three active substances were authorized against aphids on sugar beet in 2018, and all three were withdrawn by the neonicotinoid ban, so that the ratio equals one and no alternative remained in 2019. By contrast, two of the nine fungicides authorized against rust fungi on wheat were withdrawn in 2016, a ratio of 0.22, and seven alternatives remained in 2017.

Every use thus experiences a regulatory event, defined as the largest ban in its history, but these events differ in intensity. I split uses into two groups. The 291 treated uses have no alternative active substance available in the year following their regulatory event, whose intensity is therefore equal to one. The 669 control uses keep at least one active substance after their regulatory event, whose intensity is below one. For each use, treated or control, I measure time relative to its own regulatory event, and keep the years from five years before to eight years after it. Treated and control uses are thus compared at the same distance from the largest ban in their history. The estimation sample consists of the 535 uses observed over this entire window: 182 treated uses and 353 control uses.

Specification. Indexing uses by u and years by t, I estimate the following event study:

$$\mathbf{1}\{Y_{ut} > 0\} = \sum_{k \neq -1} \delta_k \, E_{u,t-k} + \mu_u + \lambda_t + \varepsilon_{ut}, \tag{8}$$

where Y_{ut} is the number of new active substances authorized for use u in year t, $E_{u,t-k}$ indicates that use u is treated and experienced its regulatory event in year $t-k$, μ_u are use fixed effects, and λ_t

are year fixed effects. The coefficient δ_k measures the change in the probability that at least one new active substance is authorized for a treated use k years after its regulatory event, relative to the year before the event and to control uses observed at the same distance from their own event, in percentage points. I focus on this extensive margin because innovation counts are zero in most use-years, which makes logarithmic specifications ill-suited (Chen and Roth, 2024).

5.2 Results

Innovation. Bans that leave a use without substitutes induce innovation (Figure 15). There is no evidence of pre-trends. The probability of innovation does not increase immediately after the event, but rises gradually from the third year onwards. It becomes statistically significant six years after the event, when it reaches 5 percentage points, and remains at a similar level two years later: treated uses are then 5 percentage points more likely than control uses to see at least one new active substance authorized in a given year. This effect is large. Among uses with at least one authorized active substance in the previous year, the probability that a new one is authorized in a given year is 13% on average, so that the effect represents about 40% of this baseline. Treated uses, moreover, start from a much lower level: before their regulatory event, the probability of innovation was 0.7% for treated uses, against 12% for control uses. Relative to their own pre-event level, the effect is therefore considerable.[18] Bans on products without substitutes thus create strong incentives for agrochemical companies to fill the void they leave.

Adaptation versus new molecules. To understand the nature of these innovations, I distinguish between two types. Adaptation-based innovations are active substances already authorized for other uses, on another crop or against another target, that are authorized for the affected use. They correspond to incremental innovations, which repurpose existing molecules and require new trials and authorization procedures. New-to-the-world innovations are molecules authorized for the first time.[19] They correspond to more radical innovations, which require the discovery and development of a new active substance.

Before their regulatory event, treated uses almost never received adapted substances or new molecules,

[18] These differences in levels are absorbed by use fixed effects, and the identifying assumption is that, absent the event, the probability of innovation of treated and control uses would have followed parallel trends. The raw probability of innovation of control uses is also higher after their own regulatory event (16%, against 12% before), so that the estimates measure the additional innovation induced by the loss of all substitutes relative to a partial ban, rather than the total effect of a ban.

[19] Since E-Phy covers products authorized in France, a new-to-the-world innovation is a molecule that had never been authorized for any use in France before.

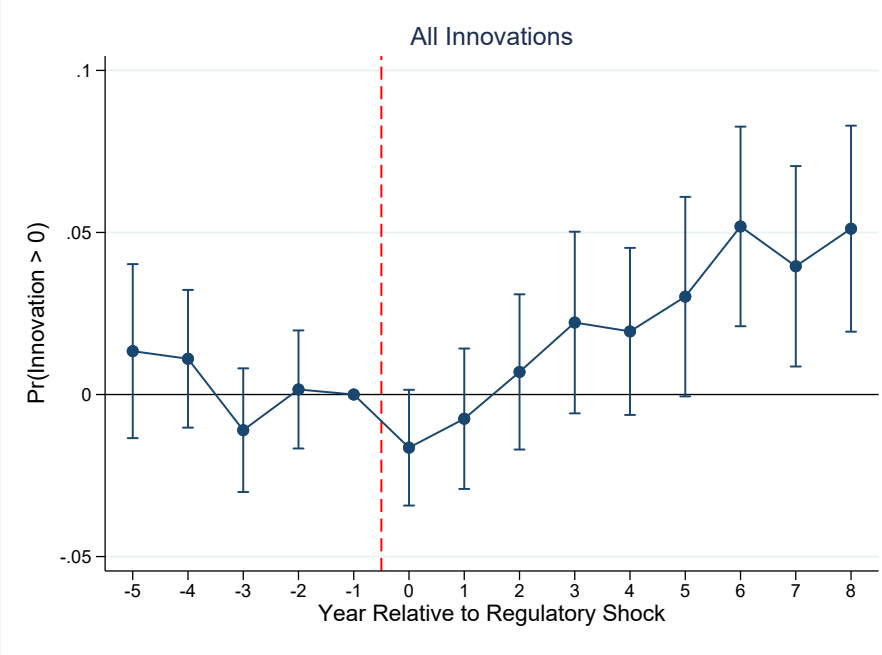


Figure 15. Effect of Pesticide Bans on Innovation

Notes: This figure reports the estimates of equation (8), along with 95% confidence intervals. The outcome is an indicator equal to one if at least one new active substance is authorized for the use in the year. Treated uses have no alternative active substance available in the year following their regulatory event, and control uses keep at least one. The specification includes use and year fixed effects. Source: E-Phy.

with annual probabilities of 0.4% and 0.2%, against 6% and 7% for control uses. Both types of innovation respond to bans without substitutes, with similar magnitudes. The probability of an adaptation-based innovation increases gradually and becomes statistically significant eight years after the event, when it reaches 3 percentage points (Figure 16). New-to-the-world innovation responds earlier: the effect becomes statistically significant after five years, reaches 3.4 percentage points after six years, and remains at 3 percentage points after eight years (Figure 17). Bans thus do not only lead firms to repurpose existing molecules: they also trigger the introduction of entirely new ones. This contrasts with the case of CFCs, where substitutes were mostly new to the industry rather than new to the world (Dugoua, 2025).

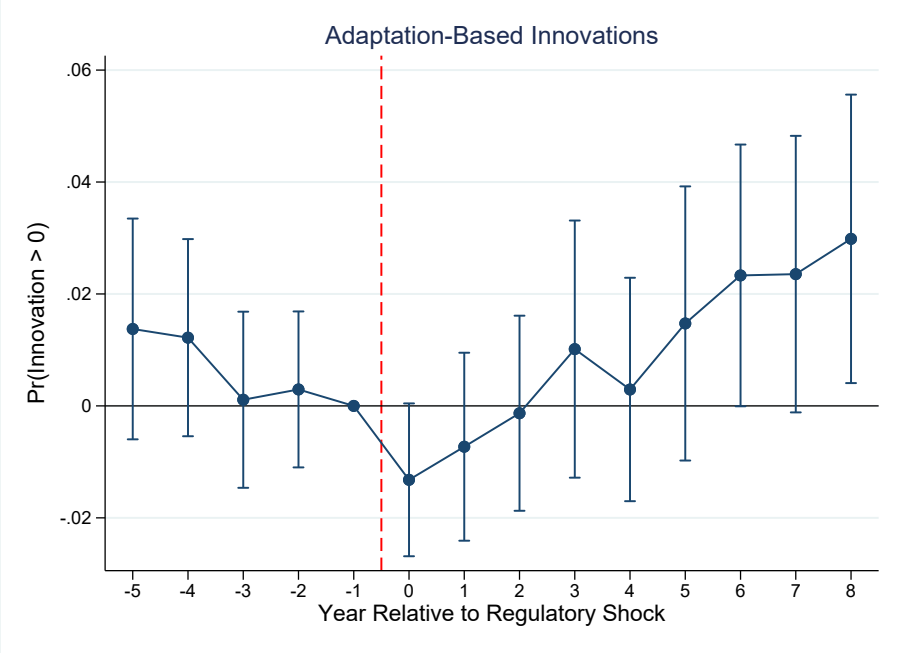


Figure 16. Effect of Pesticide Bans on Adaptation-Based Innovation

Notes: This figure reports the estimates of equation (8), along with 95% confidence intervals. The outcome is an indicator equal to one if at least one active substance already authorized for other uses is authorized for the use in the year. The sample and specification are the same as in Figure 15.

Availability of substitutes. Innovation responds to bans, but does it restore the options available to farmers? Appendix Figure A18 reports the effect of bans on the probability that at least one active substance is authorized for the use. By construction, treated uses lose all their authorized active substances in the year of the event. Their probability of having at least one then recovers only slowly, as new substances are authorized. Eight years after the event, treated uses remain 87 percentage points less likely than control uses to have at least one authorized active substance. Innovation thus replenishes only a small part of the lost arsenal in the medium term, and most uses left without substitutes

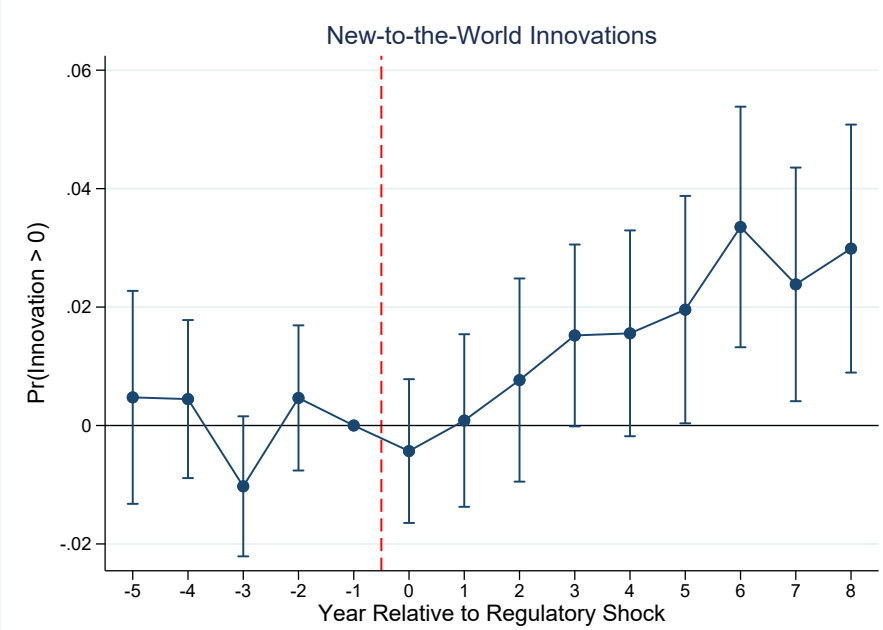


Figure 17. Effect of Pesticide Bans on New-to-the-World Innovation

Notes: This figure reports the estimates of equation (8), along with 95% confidence intervals. The outcome is an indicator equal to one if at least one active substance authorized for the first time is authorized for the use in the year. The sample and specification are the same as in Figure 15.

remain uncovered. In the meantime, farmers may have to rely on less effective methods, suffer yield losses, or abandon the crop.

Implications for regulation. The effect of bans on innovation is thus the mirror image of their effect on productivity: bans on products without substitutes are the most costly for farms, but also those that induce the most innovation. Innovation, however, takes time, and during this period farmers face constrained pest control options. Regulation therefore involves a trade-off between short-term productivity losses and medium-term technological progress. Banning products without substitutes can be justified not only on environmental grounds, but also as a way to induce the development of safer pest control methods. It should then be accompanied by measures that bridge the gap until innovation catches up, such as targeted support for research and development, incentives for integrated pest management, or temporary support for farmers.

6 Conclusion

This paper uses French farm-level data to study how bans on crop protection products affect the productivity of the farms that rely on them and innovation by the firms that supply them. The central finding is that the availability of substitutes shapes both effects, in opposite directions. Bans on products without substitutes are the most costly for farms, but also those that induce the most innovation.

The neonicotinoid ban, which removed an input without effective substitute from sugar beet cultivation, illustrates the costs of regulation for the economic agents it targets, here farms. Between 2018 and 2022, the revenue land productivity of exposed farms declined by about 0.3% per percentage point of pre-ban sugar beet land share, and their value-added land productivity declined about three times as much. The direct effect of the ban on the yield of the product line it targets accounts for about a third of this decline, and the induced reorganization of the production function for about a fifth, the remainder reflecting relative price movements on world commodity markets. Beyond these costs for the farms it targeted, the ban generated positive environmental spillovers: it raised the land productivity of neighboring orchards, whose yields depend on pollination, although its effect on neighboring farms as a whole is a precisely estimated zero.

Extending the analysis to all bans on crop protection products affecting French field crops, I find that the land productivity of exposed farms declines on average by about 3% relative to non-exposed farms five years after a ban. This average masks large differences: the decline reaches about 10% when the banned products have no substitute, is halved when substitutes are scarce, and is zero when they are abundant.

From a static perspective, these results support the principle of not banning a product without substitutes. Accounting for innovation qualifies this conclusion. Bans that leave a use without any authorized product increase the probability that a new active substance is authorized for this use by about 5 percentage points six to eight years later. This effect represents about 40% of the average annual probability of innovation, for uses that previously attracted almost none. Agrochemical firms respond both by adapting existing substances to the affected uses and by introducing entirely new molecules. This innovation, however, takes time, and during the years that follow a ban, farmers face constrained pest control options.

Regulation therefore involves a trade-off between short-term productivity losses and medium-term technological progress, with three implications for policy. First, bans on products without substitutes

impose the largest costs on farms. They can nonetheless be justified not only by the negative environmental externalities of these products, but also as a way to induce the development of new pest control solutions. Support to farms should then target the losses that the regulation causes: in the neonicotinoid case, a compensation scheme should be calibrated on the yield and reallocation channels, not on relative price movements, which are driven by world markets. Second, practices that reduce the dependence of farms on chemical inputs make them more resilient to bans, and their adoption could be encouraged ahead of regulatory changes. Third, regulation must be predictable and credible. Farms reorganize their production in response to bans, and the temporary derogation granted to sugar beet growers did not reverse this reorganization. Agrochemical firms, for their part, have stronger incentives to invest in research when they expect the demand for substitutes to last. Short-term derogations may blur these signals without restoring the situation that prevailed before the ban.

Economists have long debated the effect of regulation on productivity. The question is difficult to answer because regulation rarely targets an input that is easily measurable in firm-level data, and because many confounding factors shape a firm's production function at the same time. This paper addresses this question in a setting where the production function is comparatively simple and observable: French crop farming, in which land is the main fixed input, crops are distinct product lines, and regulation targets specific inputs for specific products. More generally, the rare combination of detailed data on inputs, outputs, and technology adoption available in agriculture makes it a natural laboratory for studying economic mechanisms that are hard to identify in standard firm-level data, an avenue that future research could pursue further.

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A Appendix Tables and Figures

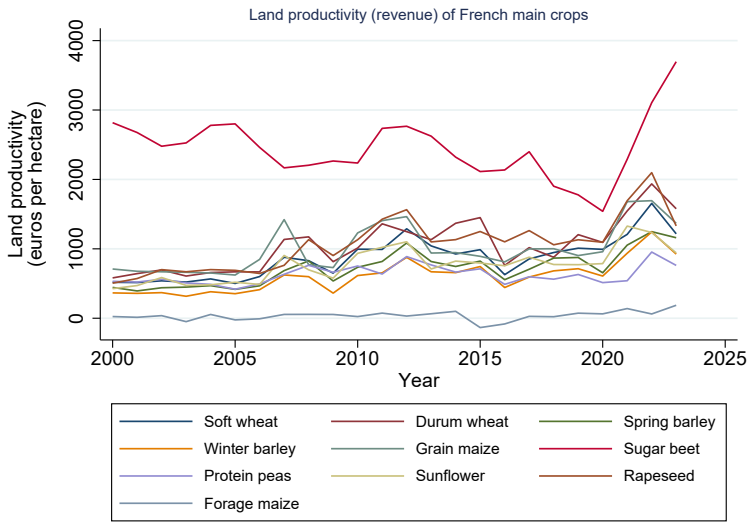


(a) Leaves of sugar beet infected with Beet Yellows Virus



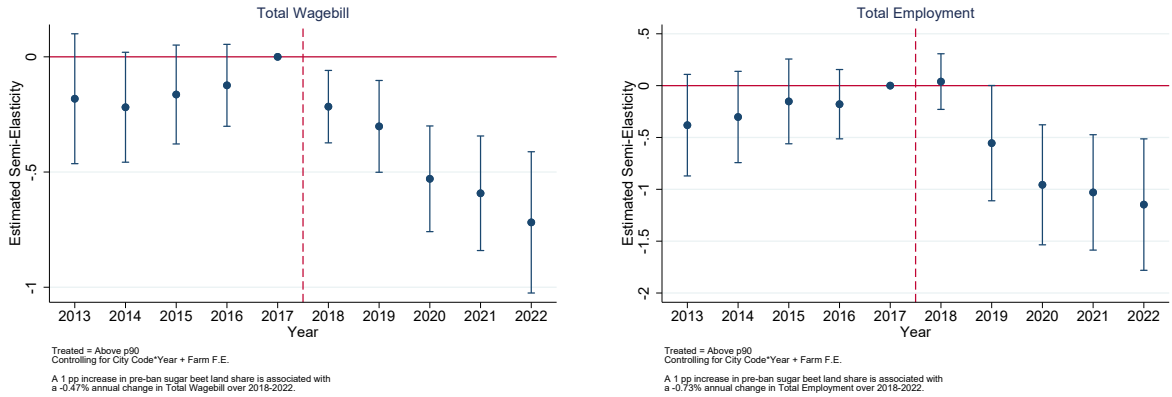
(b) Comparison of the size of a healthy sugar beet (left) and a severely affected one (right) by Beet Yellows Virus

Appendix Figure A1. Effects of Beet Yellows Virus on Sugar Beet Cultivation



Appendix Figure A2. Revenue per Hectare of the Main Crops Cultivated in France

Notes: This figure reports the average revenue per hectare of the ten main crops cultivated in France. Source: FADN.

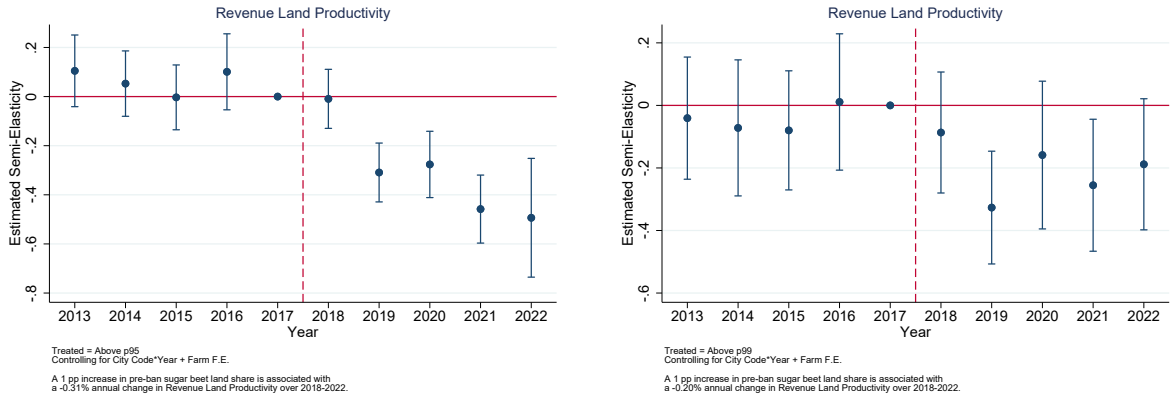


(a) Total wage bill

(b) Total employment

Appendix Figure A3. Effect of the Neonicotinoid Ban on Employment and Earnings

Notes: This figure reports the estimates of equation (1), along with 95% confidence intervals, with the log of the total wage bill (panel a) and of total employment in full-time equivalents (panel b) as outcomes. Estimates are rescaled as in Figure 3. The sample and fixed effects are the same as in Figure 3.

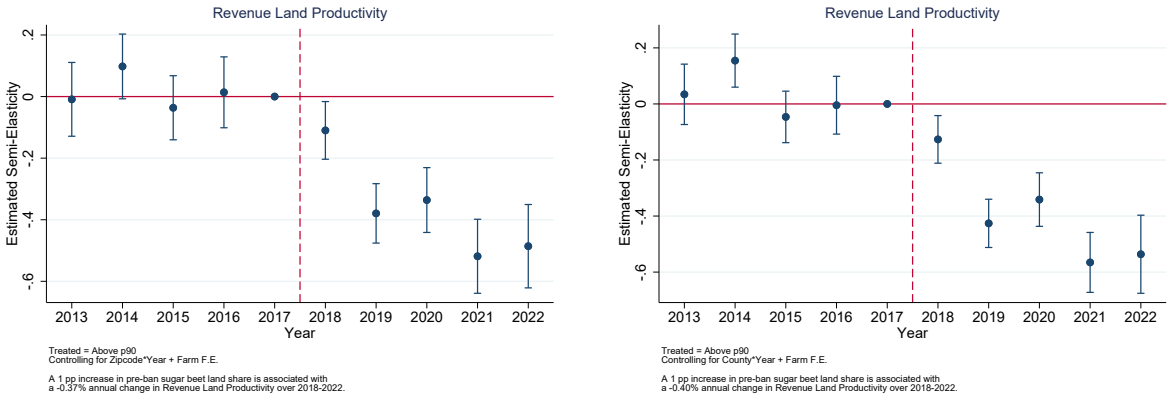


(a) 95th percentile of sugar beet land share

(b) 99th percentile of sugar beet land share

Appendix Figure A4. Neonicotinoid Ban: Alternative Exposure Thresholds

Notes: This figure reports the estimates of equation (1) with log land productivity as the outcome, along with 95% confidence intervals. Exposed farms are those whose sugar beet land share in 2015 exceeds the 95th percentile (panel a) or the 99th percentile (panel b). Estimates are rescaled as in Figure 3. The sample and fixed effects are the same as in Figure 3.

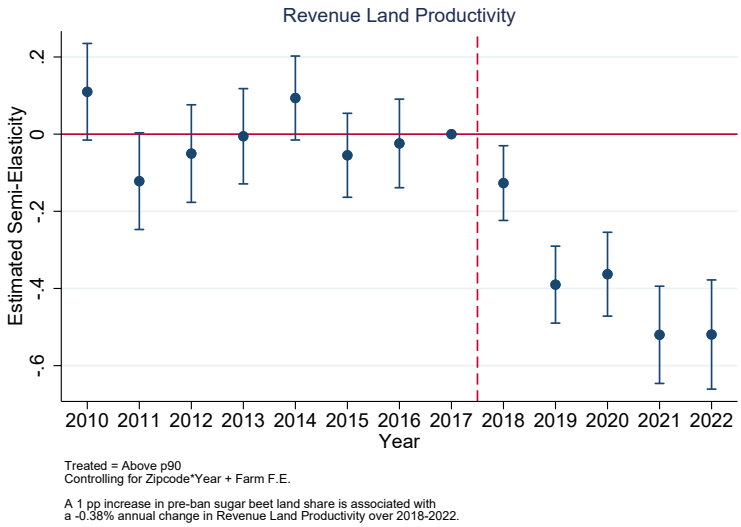


(a) Zipcode-by-year fixed effects

(b) County-by-year fixed effects

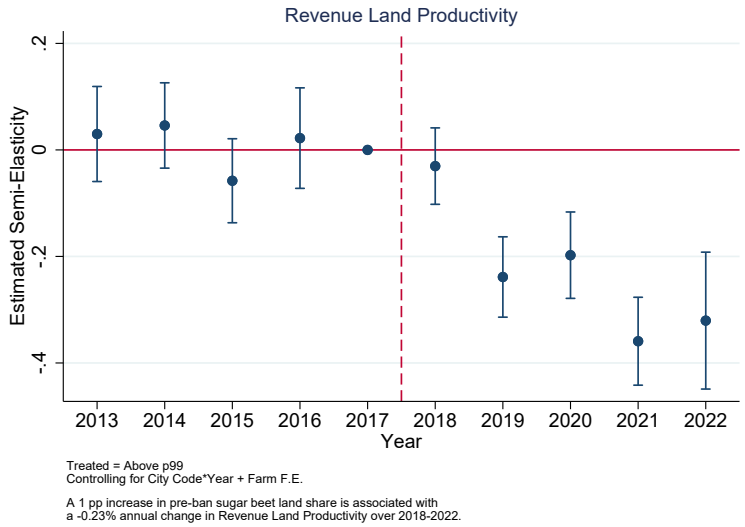
Appendix Figure A5. Neonicotinoid Ban: Alternative Fixed Effects

Notes: This figure reports the estimates of equation (1) with log land productivity as the outcome, along with 95% confidence intervals, replacing city-by-year fixed effects with zipcode-by-year fixed effects (panel a) or county-by-year fixed effects (panel b). Estimates are rescaled as in Figure 3. The sample is the same as in Figure 3.



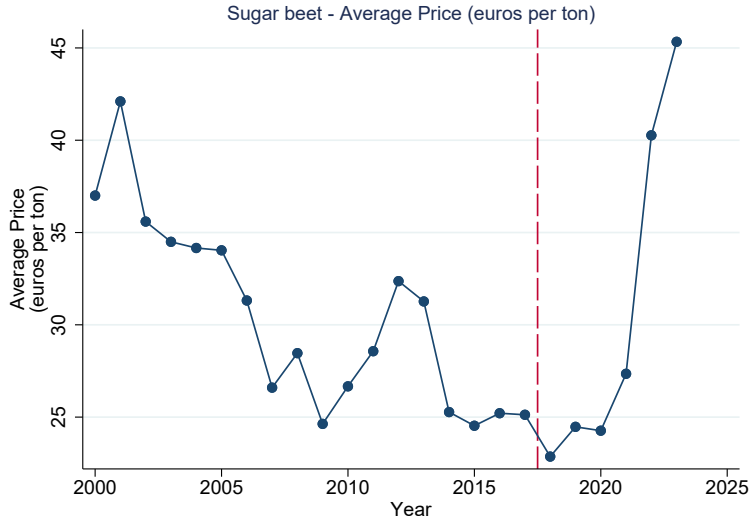
Appendix Figure A6. Neonicotinoid Ban: Panel Balanced from 2010

Notes: This figure reports the estimates of equation (1) with log land productivity as the outcome, along with 95% confidence intervals. The sample is a balanced panel of farms that grew sugar beet between 2015 and 2017, observed from 2010 to 2022. Estimates are rescaled as in Figure 3. The specification includes farm fixed effects and zipcode-by-year fixed effects.



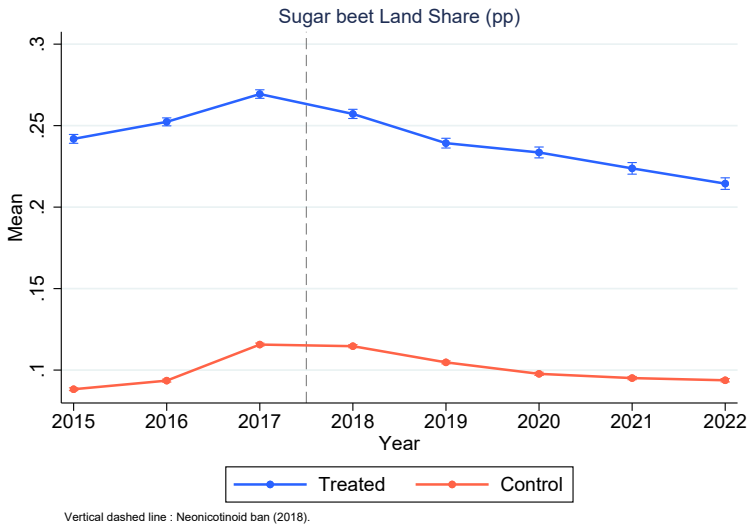
Appendix Figure A7. Neonicotinoid Ban: Extended Control Group

Notes: This figure reports the estimates of equation (1) with log land productivity as the outcome, along with 95% confidence intervals. The sample is a balanced panel of all farms observed from 2013 to 2022, regardless of whether they grew sugar beet before the ban. Exposed farms are those whose sugar beet land share in 2015 exceeds the 99th percentile (24% of cultivated land). Estimates are rescaled as in Figure 3. The specification includes farm fixed effects and city-by-year fixed effects.



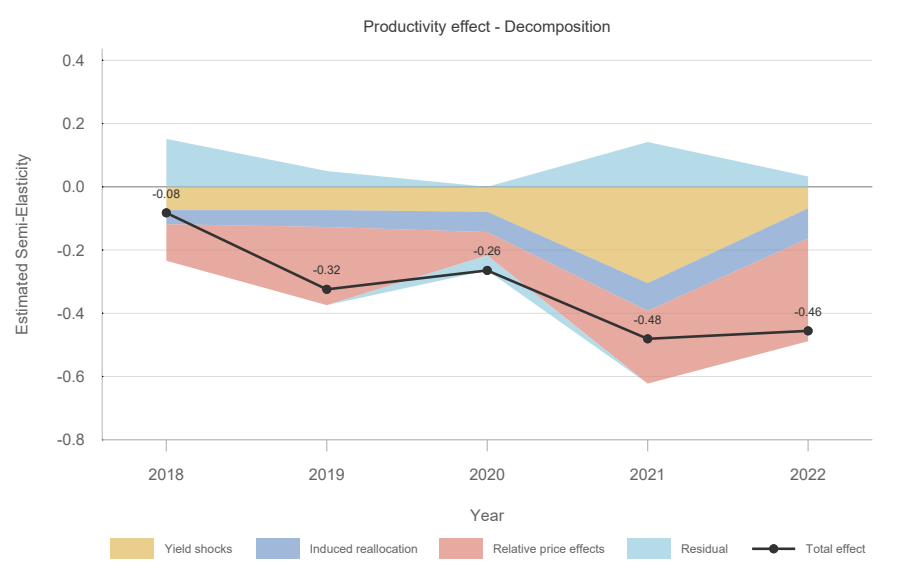
Appendix Figure A8. Average Sugar Beet Price

Notes: This figure reports the average price of sugar beet, in euros per tonne. The vertical line marks the neonicotinoid ban (2018). Source: FADN.



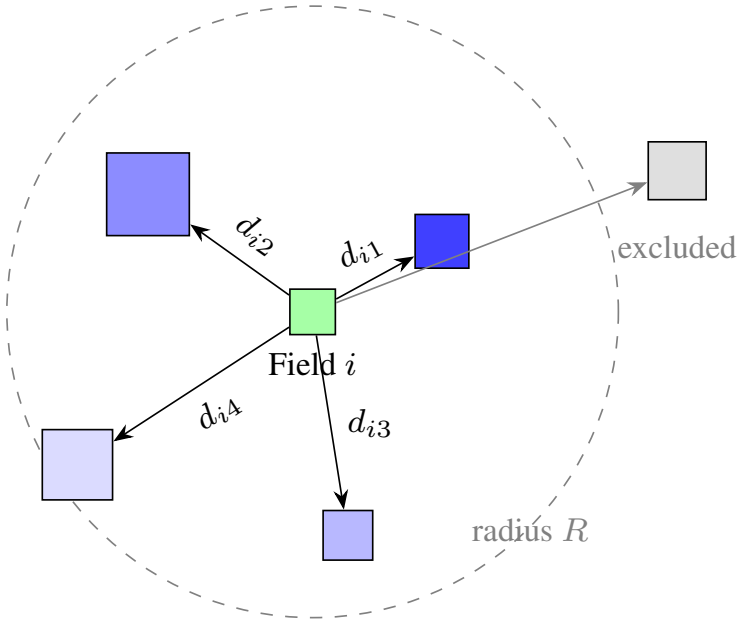
Appendix Figure A9. Sugar Beet Land Share of Exposed and Control Farms

Notes: This figure plots the average share of farm land devoted to sugar beet for exposed farms, whose sugar beet land share in 2015 exceeds the 90th percentile, and for control farms. The sample is the same as in Figure 3. The vertical line marks the neonicotinoid ban (2018). Source: Graphical Land Parcel Register.



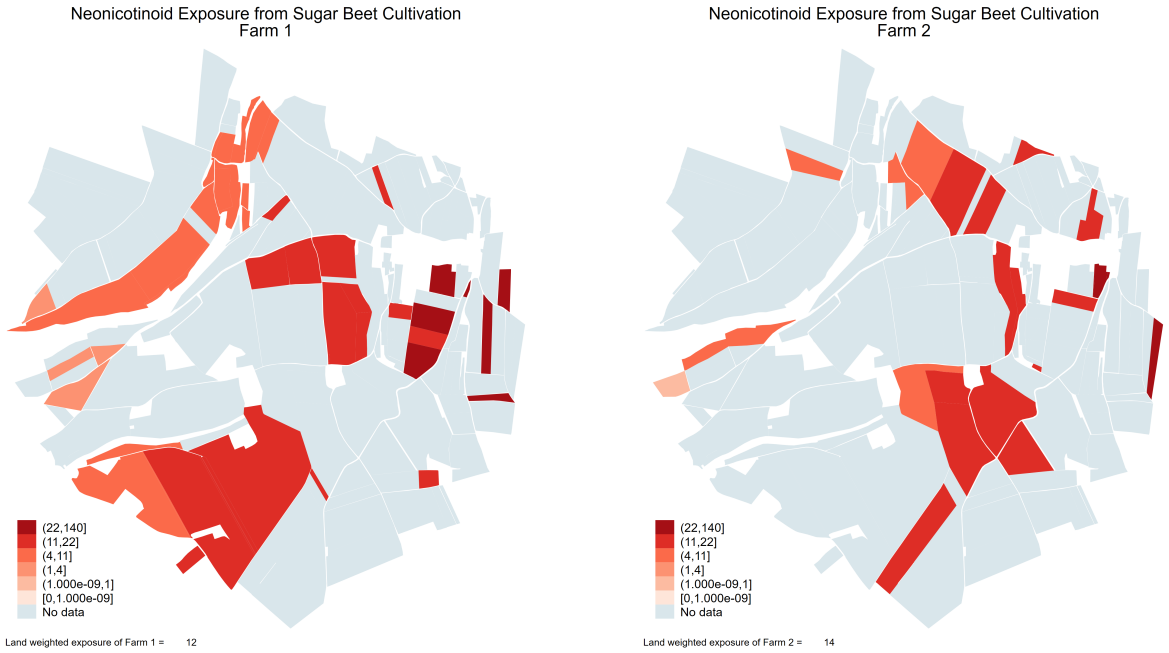
Appendix Figure A10. Decomposition of the Effect of the Neonicotinoid Ban on Land Productivity over Time

Notes: This figure decomposes the effect of the neonicotinoid ban on land productivity in each year from 2018 to 2022 into the terms of equation (3). Each term is rescaled as in Figure 7. The total effect corresponds to the baseline estimates of Figure 3. The terms of the decomposition are computed on the farms from the main sample that are also surveyed in the FADN.



Appendix Figure A11. Construction of the Local Exposure Index

Notes: This figure illustrates the construction of the exposure of parcel i to sugar beet cultivation, defined in equation (4). Each sugar beet parcel j located within a radius $R = 1{,}500$ metres contributes its area a_j, weighted by $e^{-d_{ij}/\rho}$, where d_{ij} is the distance between the centroids of the two parcels and $\rho = 600$ metres.

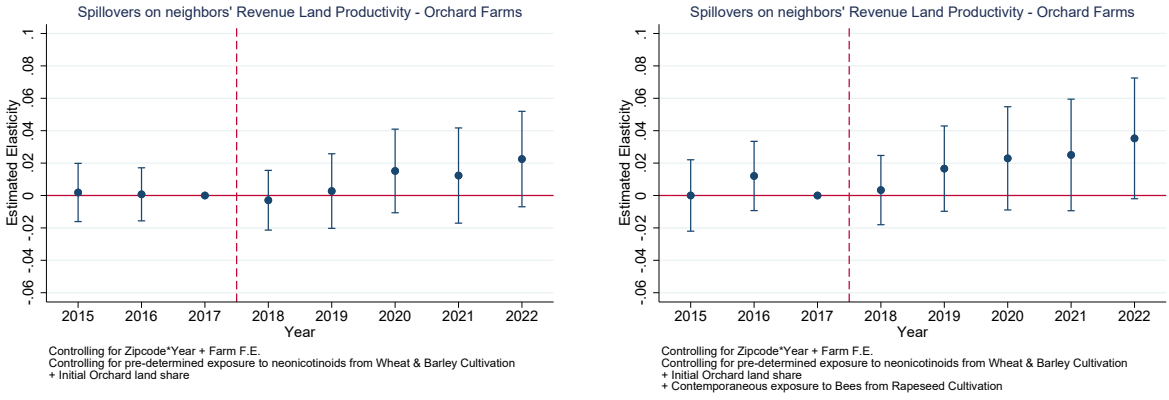


(a) Farm 1: lower exposure

(b) Farm 2: higher exposure

Appendix Figure A12. Local Exposure of Two Fictitious Farms Located in the Same Zipcode

Notes: This figure shows two fictitious farms, used for illustration only, located in the same zipcode, together with the exposure of surrounding parcels to sugar beet cultivation in 2015, as defined in equation (4). Farm 2 is more exposed to neonicotinoids than farm 1.

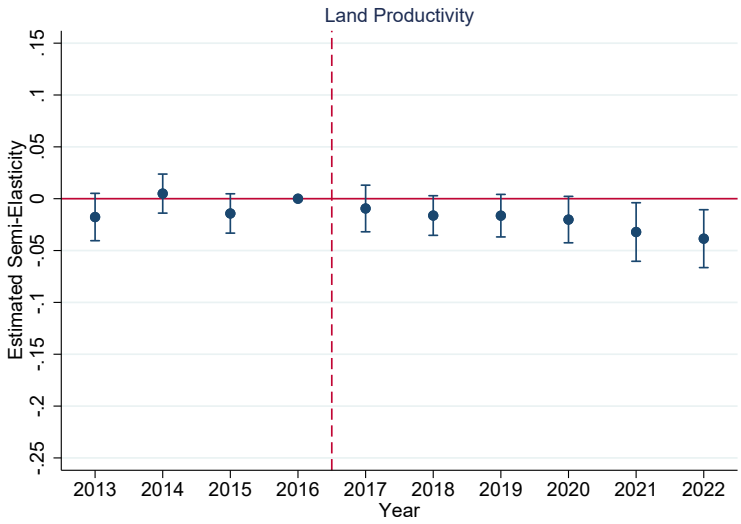


(a) Controlling for the initial orchard land share

(b) Adding contemporaneous exposure to rapeseed

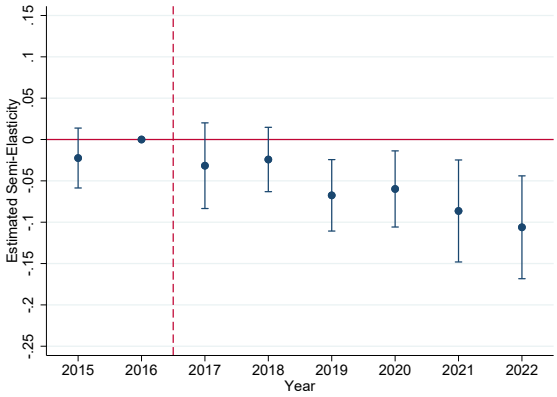
Appendix Figure A13. Spillovers on Orchards: Additional Controls

Notes: This figure reports the estimates of β_k^{beet} in equation (5) for orchard farms, along with 95% confidence intervals. Panel (a) adds the initial orchard land share, interacted with year indicators, as a control. Panel (b) further adds contemporaneous exposure to rapeseed cultivation, constructed as in equation (4). All specifications control for pre-ban exposure to wheat and barley cultivation and include farm fixed effects and zipcode-by-year fixed effects.

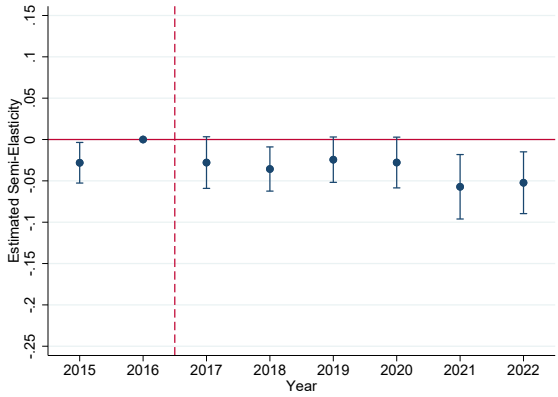


Appendix Figure A14. Pesticide Bans: Less Stringent Sample Selection

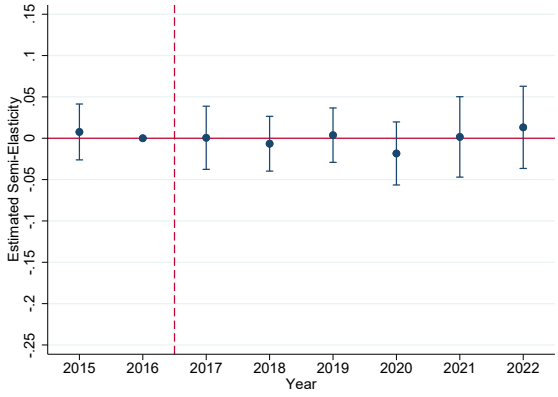
Notes: This figure reports the estimates of equation (6) for log land productivity, with 95% confidence intervals, without the condition that the surveyed crop represents at least 10% of the farm's cultivated land. The specification is the same as in Figure 13.



(a) No substitute available



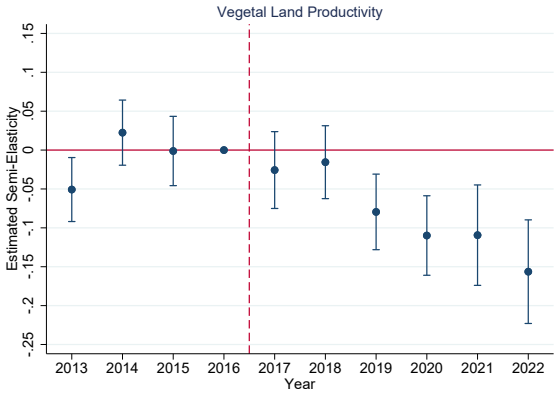
(b) Number of substitutes below the median



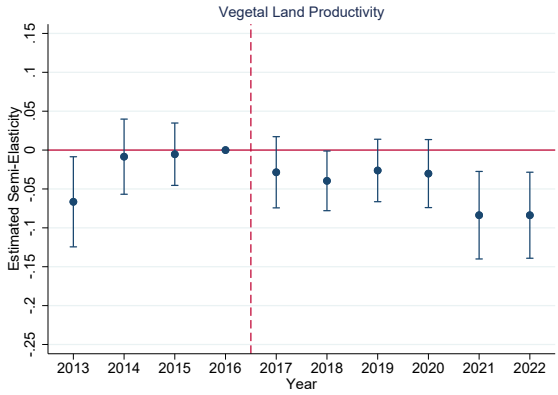
(c) Number of substitutes above the median

Appendix Figure A15. Pesticide Bans by Number of Substitutes: CAP-Declared Land as Denominator

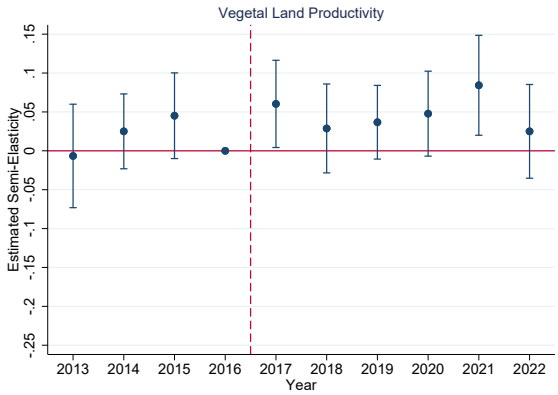
Notes: This figure reproduces Figure 14, measuring land productivity as total revenues excluding subsidies divided by the cultivated land declared under the Common Agricultural Policy, available from 2015.



(a) No substitute available



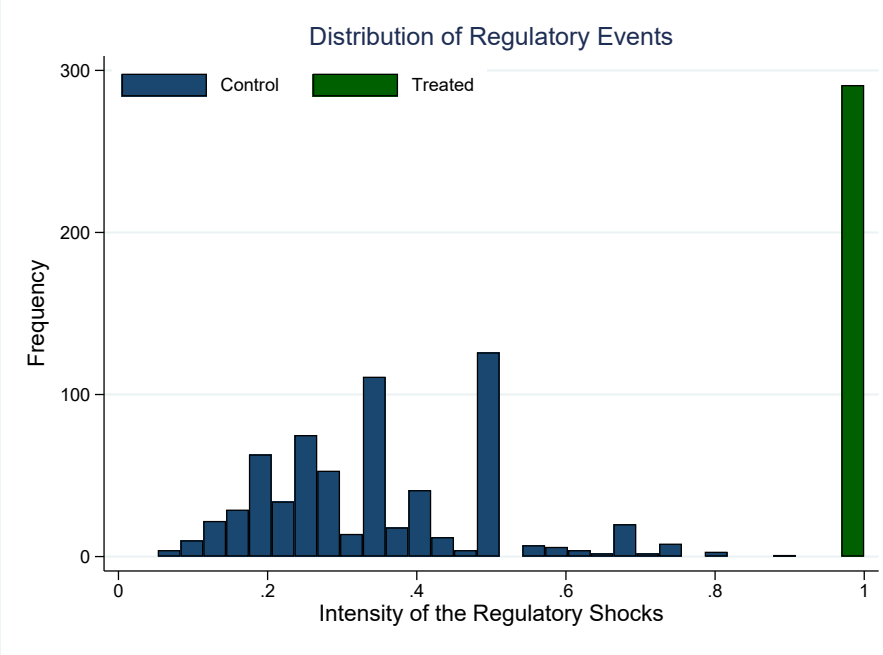
(b) Number of substitutes below the median



(c) Number of substitutes above the median

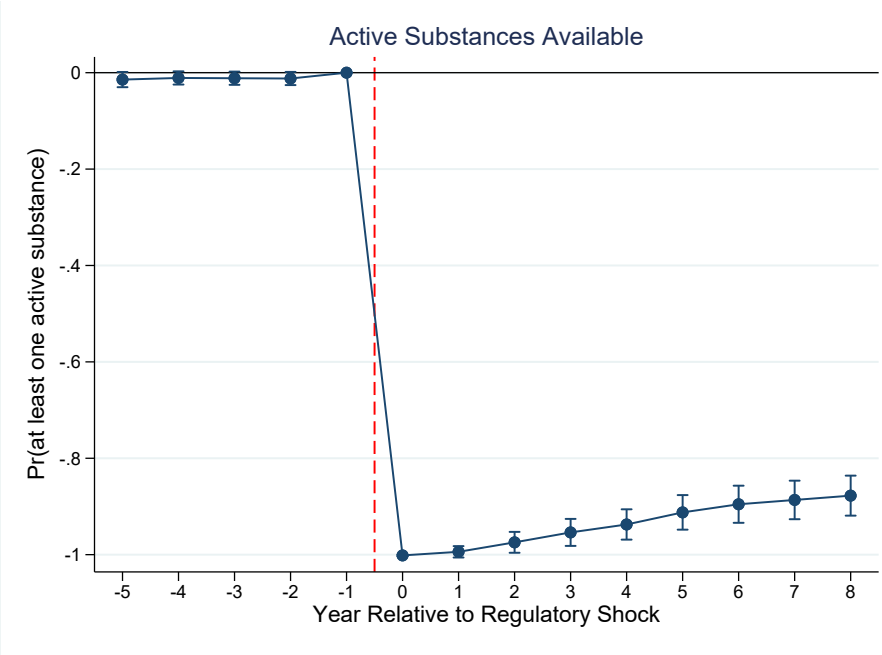
Appendix Figure A16. Pesticide Bans by Number of Substitutes: Crop Revenues as Numerator

Notes: This figure reproduces Figure 14, measuring land productivity as crop production revenues, instead of total revenues excluding subsidies, per hectare of cultivated land.



Appendix Figure A17. Distribution of the Intensity of Regulatory Events

Notes: This figure reports the distribution across crop–target uses of the intensity of their regulatory event, R_{u,τ_u}, defined in equation (7) as the largest share of the products available for the use that are banned in a single year. Treated uses have no alternative active substance available in the year following their regulatory event, and control uses keep at least one. Source: E-Phy.



Appendix Figure A18. Effect of Pesticide Bans on the Availability of Active Substances

Notes: This figure reports the estimates of equation (8), along with 95% confidence intervals. The outcome is an indicator equal to one if at least one active substance is authorized for the use in the year. The sample and specification are the same as in Figure 15.

	Treated	Control
Cultivated land (ha)	159	149
Sugar Beet Land Share (%)	24	9
Production (k€)	348	316
Crop production (k€)	297	230
Subsidies (k€)	51	47
Net stock of machines (k€)	140	145
Land productivity (k€ per ha)	2.31	2.25
Employment (FTE)	1.19	1.18
Number of farms	1,560	14,042

Appendix Table A1. Characteristics of Farms Exposed and Not Exposed to the Neonicotinoid Ban

Notes: This table presents descriptive statistics for the two groups in 2015, three years before the implementation of the Neonicotinoid Ban.

	Exposed	Non-exposed
Cultivated land (ha)	170	153
Production (k€)	332	289
Crop production (k€)	241	167
Subsidies (k€)	52	46
Net stock of machines (k€)	168	146
Land productivity (k€ per ha)	2.19	2.04
Employment (FTE)	1.18	1.11
Number of farms	863	2,304

Appendix Table A2. Characteristics of Farms Exposed and Not Exposed to a Pesticide Ban

Notes: This table presents descriptive statistics for the two groups in 2015, two years before the year of the studied pesticide ban.

B Decomposition of the Effect on Land Productivity

This appendix derives the decomposition of the effect of the neonicotinoid ban on land productivity presented in equation (3). The derivation proceeds in three steps. First, I show that the event-study coefficient, estimated in logs, can be approximated by a difference-in-differences in levels, normalized by pre-ban land productivity. Second, I write an accounting identity for the gap in land productivity between exposed and control farms. Third, I take the difference-in-differences of this identity and isolate the contribution of each channel. Table B1 summarizes the notation.

Appendix Table B1. Notation

Variable	Definition	Unit
Y_{it}	Land productivity of farm i in year t (revenue per hectare)	EUR/ha
$\bar{Y}_0$	Average land productivity of the two groups in 2017	EUR/ha
s^b_{it}	Share of cultivated land devoted to sugar beet	fraction
q_t	Sugar beet price, common to all farms	EUR/t
y^b_{it}	Sugar beet yield	t/ha
p^o_{it}	Revenue per hectare of crops other than sugar beet	EUR/ha
$\bar{\pi}_t$	Profitability gap of sugar beet: $\bar{\pi}_t = q_t\,\bar{y}^b_t - \bar{p}^o_t$	EUR/ha
Δs^b_t	Change in the sugar beet share gap since 2017: $\Delta s^b_t = \tilde{s}_t - \tilde{s}_0$	fraction
$\Delta\bar{\pi}_t$	Change in the profitability gap since 2017: $\Delta\bar{\pi}_t = \bar{\pi}_t - \bar{\pi}_0$	EUR/ha
$\delta\bar{y}^b_t,\ \delta q_t,\ \delta\bar{p}^o_t$	Changes since 2017 in the average sugar beet yield, the sugar beet price, and the average revenue per hectare of other crops	

Notes: Superscripts T and C denote exposed and control farms. For any variable x, $\bar{x} = (x^T + x^C)/2$ denotes the average of the two groups and $\tilde{x} = x^T - x^C$ the gap between them. Subscript 0 denotes the reference year 2017, the last year before the ban.

B.1 From the Log Estimand to a Decomposition in Levels

Denote by $\bar{\gamma}_t$ the coefficient δ_t of equation (1) for log land productivity, estimated on the FADN subsample. Abstracting from fixed effects, it is a difference-in-differences in logs:

$$\bar{\gamma}_t = \big(E[\log Y^T_{it}] - E[\log Y^T_{i0}]\big) - \big(E[\log Y^C_{it}] - E[\log Y^C_{i0}]\big)\,.$$

The decomposition, by contrast, is naturally written in levels, since land productivity is a sum across crops. Three approximations link the two.

Log approximation. For each farm, $\log Y_{it} - \log Y_{i0} \approx (Y_{it} - Y_{i0})/Y_{i0}$. This first-order approxi-

mation is accurate when changes in land productivity are small, which is the case here: the estimated effects are around -5%.

Homogeneous baselines within groups. Within a group g, the average growth rate $E^g[(Y_{it} - Y_{i0})/Y_{i0}]$ equals $E^g[Y_{it} - Y_{i0}]/\bar{Y}_0^g$ plus a covariance term between the change in land productivity and the inverse of its baseline level. I neglect this term, which is small when, within each group, the change in land productivity is roughly independent of its baseline level. Since farm fixed effects absorb the time-invariant component of land productivity, this assumption only concerns its residual variation.

Comparable baselines across groups. If exposed and control farms have similar pre-ban land productivity, $\bar{Y}_0^T \approx \bar{Y}_0^C \approx \bar{Y}_0$, the growth rates of the two groups share a common denominator. This holds in the data: although exposed farms devote 24% of their land to sugar beet against 9% for control farms, their pre-ban land productivity is only 2.1% higher.

Under these three approximations,

$$\bar{\gamma}_t \approx \frac{\Delta Y_t}{\bar{Y}_0}, \qquad \text{with} \quad \Delta Y_t = \left(\bar{Y}_t^T - \bar{Y}_t^C\right) - \left(\bar{Y}_0^T - \bar{Y}_0^C\right). \tag{9}$$

Any additive decomposition of the difference-in-differences in levels, $\Delta Y_t = \sum_k T_k$, therefore translates into contributions $T_k/\bar{Y}_0$ to the event-study coefficient.

B.2 An Accounting Identity

Land productivity in levels. Distinguishing sugar beet (b) from all other crops (o), the land productivity of a farm is the average of the revenue per hectare of each, weighted by their land shares:

$$Y_{it} = s_{it}^b \, q_t \, y_{it}^b + \left(1 - s_{it}^b\right) p_{it}^o, \tag{10}$$

where the revenue per hectare of sugar beet is the product of its price and yield. Sugar beet growers sell their production to processors, mostly cooperatives, and take its price as given. I therefore treat the sugar beet price q_t as common to all farms.

Gap between exposed and control farms. I average equation (10) within each group and use the fact that, for any two variables a and b, $a^T b^T - a^C b^C = \bar{a}\,\tilde{b} + \bar{b}\,\tilde{a}$. This yields a decomposition of the

land productivity gap between the two groups:[20]

$$\bar{Y}_t^T - \bar{Y}_t^C = \underbrace{q_t\,\bar{s}_t^b\,\tilde{y}_t}_{\text{yield gap}} + \underbrace{\tilde{s}_t\,\bar{\pi}_t}_{\text{crop mix}} + \underbrace{\left(1-\bar{s}_t^b\right)\tilde{p}_t^o}_{\text{other crops}}. \tag{11}$$

The first term reflects differences in sugar beet yields between the two groups. The second reflects their different crop mix: exposed farms devote $\tilde{s}_t$ more land to sugar beet, whose revenue per hectare exceeds that of other crops by $\bar{\pi}_t$. The third reflects differences in the revenue per hectare of other crops, for instance because the two groups grow different crops.

B.3 Difference-in-Differences and Decomposition

Taking differences. The difference-in-differences in levels, ΔY_t, is the change in the gap (11) between 2017 and year t. I assume that, in 2017, the two groups have similar sugar beet yields and similar revenue per hectare on other crops, $\tilde{y}_0 = 0$ and $\tilde{p}_0^o = 0$, two assumptions that can be checked in the pre-ban data. Replacing $\bar{s}_t^b$ by $\bar{s}_0^b$, which only neglects second-order terms, gives

$$\Delta Y_t \approx q_t\,\bar{s}_0^b\,\tilde{y}_t + \left(\tilde{s}_t\,\bar{\pi}_t - \tilde{s}_0\,\bar{\pi}_0\right) + \left(1-\bar{s}_0^b\right)\tilde{p}_t^o. \tag{12}$$

Isolating the reallocation channel. The middle term combines changes in the crop mix and in the profitability of sugar beet. Adding and subtracting $\tilde{s}_t\,\bar{\pi}_0$ separates the two:

$$\tilde{s}_t\,\bar{\pi}_t - \tilde{s}_0\,\bar{\pi}_0 = \bar{\pi}_0\,\Delta s_t^b + \tilde{s}_0\,\Delta\bar{\pi}_t + \Delta s_t^b\,\Delta\bar{\pi}_t.$$

The first term values the reallocation of land away from sugar beet at the pre-ban profitability gap $\bar{\pi}_0$. Since $\bar{\pi}_0$ is measured before both the ban and the abolition of sugar quotas, the weight given to reallocation is not affected by either event.[21] The second term captures the greater exposure of farms that grow more sugar beet to changes in its profitability relative to other crops. The last term is the product of two changes and is therefore of second order.

Yields versus prices. To first order, the change in the profitability gap splits into a change in sugar

[20] This step neglects the within-group covariance between sugar beet shares and yields, $\mathrm{Cov}_g(s_{it}^b, y_{it}^b)$. This covariance is likely positive, since farms with the most suitable soils may both grow more sugar beet and obtain higher yields. Farm fixed effects, however, absorb its time-invariant component. The decomposition would only be biased if this covariance changed after the ban differently in the two groups, which is a second-order concern.

[21] Alternatively, adding and subtracting $\tilde{s}_0\,\bar{\pi}_t$ values reallocation at the current profitability gap $\bar{\pi}_t$ and absorbs the interaction term. However, $\bar{\pi}_t$ may itself respond to the ban and to the abolition of quotas, which makes this alternative less suitable.

beet yields, a change in the sugar beet price, and a change in the revenue per hectare of other crops:

$$\Delta\bar{\pi}_t \approx q_0\,\delta\bar{y}_t^b + \bar{y}_0^b\,\delta q_t - \delta\bar{p}_t^o.$$

Result. Combining these steps and dividing by $\bar{Y}_0$, following equation (9), yields equation (3) of the main text:

$$\bar{\gamma}_t \approx \frac{1}{\bar{Y}_0}\Big[\underbrace{\tilde{s}_0\,q_0\,\delta\bar{y}_t^b}_{\text{Yield shocks}} + \underbrace{\bar{\pi}_0\,\Delta s_t^b}_{\text{Induced reallocation}} + \underbrace{\tilde{s}_0\big(\bar{y}_0^b\,\delta q_t - \delta\bar{p}_t^o\big)}_{\text{Price effects}} + R_t\Big],$$

where the residual collects the remaining terms:

$$R_t = \underbrace{q_t\,\bar{s}_0^b\,\tilde{y}_t}_{\text{differential yields}} + \underbrace{\big(1-\bar{s}_0^b\big)\,\tilde{p}_t^o}_{\text{other crops}} + \underbrace{\Delta s_t^b\,\Delta\bar{\pi}_t}_{\text{interaction}}.$$

Table B2 summarizes the interpretation of each term.

Appendix Table B2. Terms of the Decomposition

Term	Expression	Interpretation
Yield shocks	$\tilde{s}_0\,q_0\,\delta\bar{y}_t^b$	Exposed farms grow more sugar beet ($\tilde{s}_0 > 0$) and are therefore more exposed to the common decline in sugar beet yields, due in particular to beet yellows virus.
Induced reallocation	$\bar{\pi}_0\,\Delta s_t^b$	Exposed farms reduce their sugar beet share more ($\Delta s_t^b < 0$) and shift land toward crops that were less profitable before the ban ($\bar{\pi}_0 > 0$).
Price effects	$\tilde{s}_0\big(\bar{y}_0^b\,\delta q_t - \delta\bar{p}_t^o\big)$	Exposed farms are more exposed to changes in the sugar beet price relative to the revenue per hectare of other crops.
Residual R_t		
Differential yields	$q_t\,\bar{s}_0^b\,\tilde{y}_t$	Change in the sugar beet yields of exposed farms relative to control farms, beyond the common change.
Other crops	$\big(1-\bar{s}_0^b\big)\,\tilde{p}_t^o$	Differential change in the revenue per hectare of other crops, for instance because the two groups replace sugar beet with different crops.
Interaction	$\Delta s_t^b\,\Delta\bar{\pi}_t$	Interaction between reallocation and the change in the profitability gap. Second order.

Summary of assumptions. The decomposition relies on four sets of assumptions: (i) the three approximations linking the event-study coefficient in logs to the difference-in-differences in levels; (ii) a sugar beet price common to all farms; (iii) no pre-ban gap between the two groups in sugar beet yields and in the revenue per hectare of other crops; and (iv) no differential change after the ban in the within-group covariance between sugar beet shares and yields.

C Mitigation Practices and the Effects of Pesticide Bans

This appendix examines whether the practices that farms had adopted to reduce their use of crop protection products before the bans mitigated the effects of these bans on their land productivity. Section 4.2 summarizes the results.

Motivation. When farmers face an abrupt change in their production conditions, such as the ban of a crop protection product, adaptation can be costly: yield losses, additional labor, investments in alternative methods, and the cost of learning new practices. Practices that reduce dependence on chemical inputs could lower these costs. Crop rotation and biological control reduce pest pressure, precision spraying and resistant varieties make it possible to use fewer inputs, and mechanical weeding can replace herbicides. Farms that had already adopted such practices before a ban may therefore be better prepared to absorb it.

Measuring mitigation practices. The Field Crop Cultural Practices Survey includes a section on the practices implemented to reduce the use of crop protection products. It records the techniques used and the reasons for adopting them, distinguishing between three motivations: health, to reduce risks for users of crop protection products or for consumers; the environment, to protect water quality and biodiversity; and economic reasons, to reduce production costs. Respondents also indicate their main motivation.

Since the bans reduce land productivity, I focus on practices adopted to limit economic losses. An exposed farm is classified as implementing mitigation practices if two conditions hold. First, its main motivation for reducing the use of crop protection products is economic. Second, it reports that at least one of the two other motivations, health or the environment, played no role in this decision. The second condition excludes farms that are equally motivated by the three reasons. Among the 863 exposed farms, 118 implemented mitigation practices according to this definition, and 745 did not.

Specification. The control group is the same as in Section 4.1, but the exposed farms are split into two groups, whose effects are estimated in a single regression:

$$\begin{aligned}\log Y_{it} = &\sum_{k \neq 2016} \delta_k^{M} \, Mitigation_i \times \mathbf{1}\{t = k\} \\ &+ \sum_{k \neq 2016} \delta_k^{NM} \, NoMitigation_i \times \mathbf{1}\{t = k\} + \mu_i + \lambda_{dt} + \varepsilon_{it},\end{aligned} \tag{13}$$

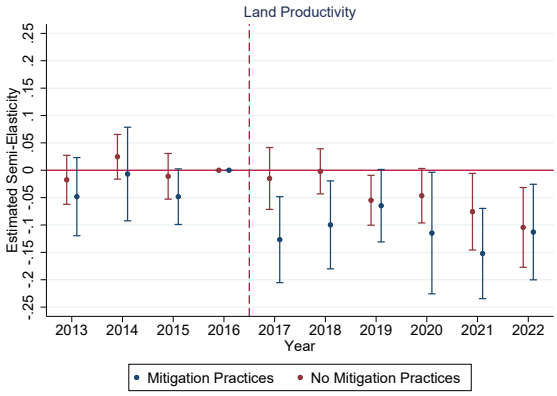
where the indicator $Mitigation_i$ identifies the exposed farms that implemented mitigation practices, the indicator $NoMitigation_i$ identifies the other exposed farms, and the remaining terms are defined as in equation (6). Since both groups share the same fixed effects, the coefficients δ_k^M and δ_k^{NM} can be compared directly.

Interpretation. Mitigation practices are chosen by farms, not randomly assigned. Farms that adopt them may differ from other exposed farms along dimensions that also shape their response to a ban, such as management quality. The estimates below should therefore be read as evidence of heterogeneity in the effect of the bans across farms, rather than as the causal effect of mitigation practices.

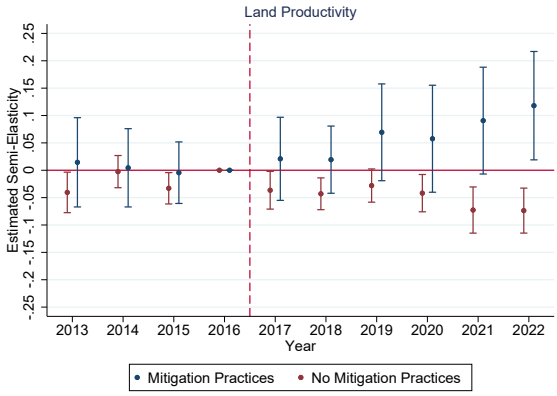
Results. Figure C1 reports the estimates of equation (13) separately by availability of substitutes for the banned products, as in Section 4.2. When no substitute is available, mitigation practices do not soften the effect of the ban (panel a). The land productivity of the two groups declines by similar amounts, by 11% for farms with mitigation practices and 10% for other farms in 2022.

When substitutes exist but are scarce, mitigation practices make a large difference (panel b). Exposed farms without such practices experience a decline in land productivity of about 7% relative to control farms in 2022. By contrast, the land productivity of exposed farms with mitigation practices increases by about 12% over the same period. The confidence intervals of the two estimates do not overlap in 2022, so that the two effects are statistically different at the 5% level. Finally, when substitutes are abundant, the ban has no effect on land productivity in either group (panel c).

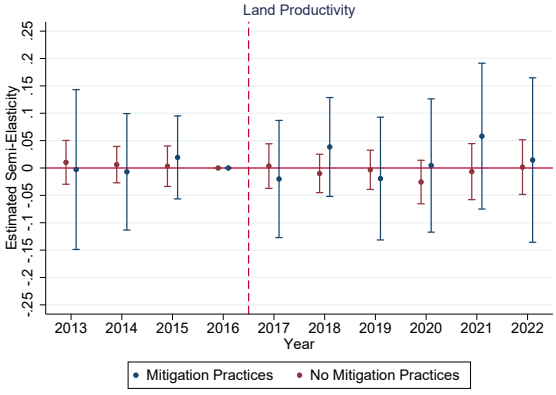
These results show that mitigation practices can offset, and even reverse, the adverse effects of a ban on productivity, but only when substitutes are available. When regulation targets products without substitutes, mitigation practices are ineffective, and the ban reduces productivity regardless of farms' preparation. Regulation can thus be compatible with farm productivity provided that it does not target irreplaceable products and that farms have prepared the transition.



(a) No substitute available



(b) Number of substitutes below the median



(c) Number of substitutes above the median

Appendix Figure C1. Effect of Pesticide Bans on Land Productivity, by Mitigation Practices and Number of Substitutes

Notes: This figure reports the estimates of equation (13) for log land productivity, with 95% confidence intervals, for exposed farms that implemented mitigation practices for economic reasons (in blue) and for other exposed farms (in red). Panels distinguish farms whose banned products had no substitute (panel a), a number of substitutes below the median of 52 (panel b), or above the median (panel c). The control group consists of farms not exposed to any ban on an economically relevant crop. The specification includes farm and county-by-year fixed effects.